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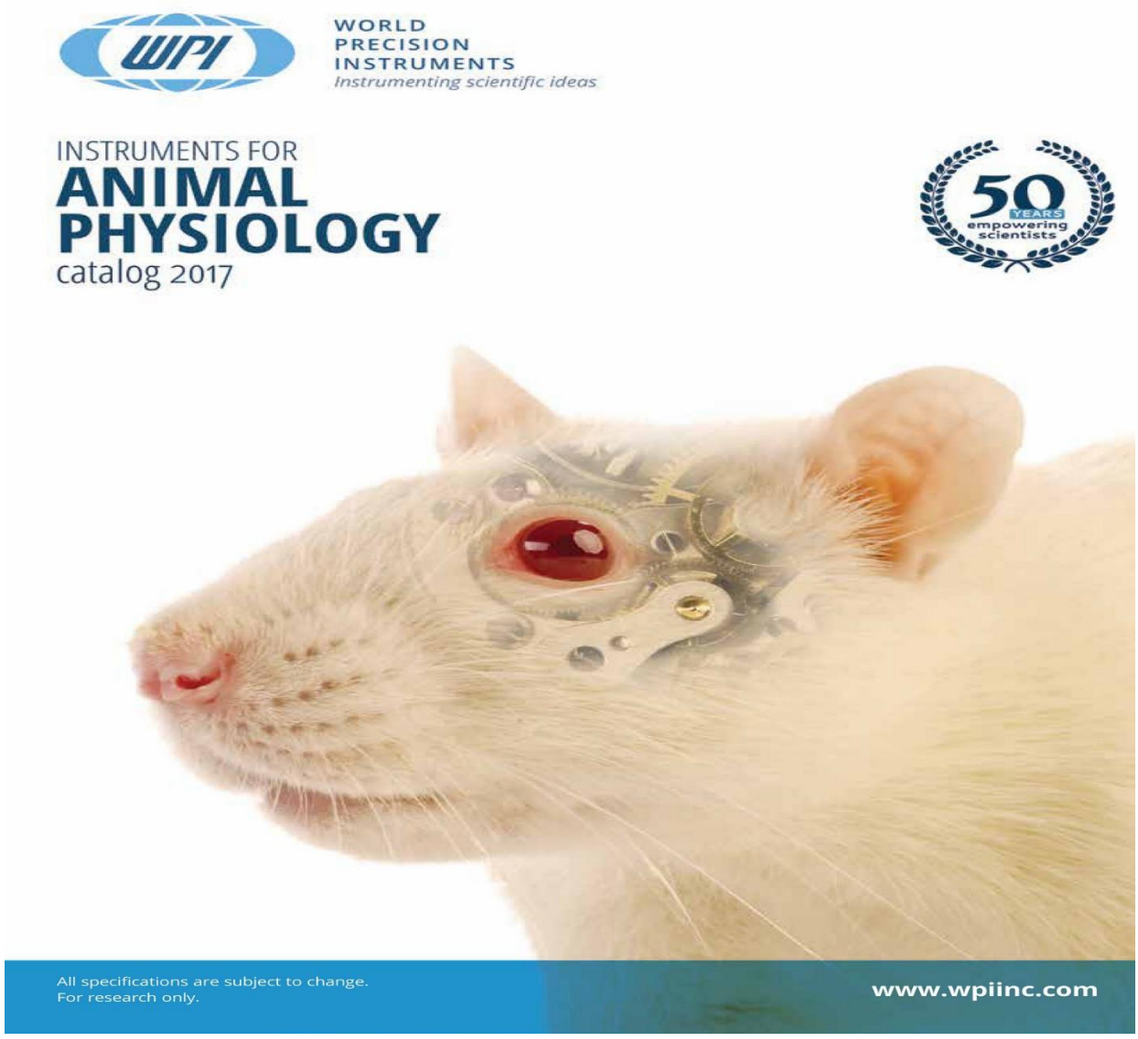
Federal Government University

MSC-ZOOLOGY

PAST SOLVED UPDATED PAPER 2022

ZOO502-Animal-Physiology-and-Behavior

QUIZ 1	QUIZ 2	QUIZ 3	GRAND QUIZ	MID TERM	FINAL TERM



MUHAMMAD IMRAN

Quiz NO.1

1. The genes for ---------- cone pigments are closely linked on the X chromosome_____ _____**Red and green**
2. Voltage gated Na+ channels are----------_____**Fast-acting channels**
3. Which of the following is an example of Apocrine Secretion?._____**Mammary glands**
4. Adenohypophysis contains ---------- types of glandular cells that synthesize and secrete six hormones_____ **Five**
5. A color-blind person who lacks ---------- cones is called a deuteranope_____ **Green**
6. Oxytocin stimulates motility of the oviduct in _____ **Birds**
7. Parathormone over secretion leads to _____**Softness of bones**
8. Substance that is released via a duct through the gland to the external or internal epithelial surface Is called as: _____**Exocrine secretion**
9. Amine hormones (thyroid and adrenal medullary hormones) are synthesized by the action of enzymes on amino acid_____**Tyrosine**
10. Salivary glands produce saliva that is delivered to the oral cavity for partial digestion of food through ---------- ducts…_____ **Parotid and submandibular**
11. Which of the following hormone acts as enteric neurotransmitter _____**Substance P**
12. Which of the following hormone Inhibits release of growth hormone? _____**Somatostatin**
13. Pituitary gland release ---------- hormones_____**9**
14. Which of the following is an example of steroid hormone? _____**Aldosterone**
15. In many organisms. ACTH is also related to the _____**Circadian rhythms**
16. That type of secretion In Which entire cell is ruptured and breaks up to release its contents is called as_____**Holocrine secretion**
17. Which of the following horm**one is released from anteri**or lobe of pituitary?. _____**Luteinizing**
18. Glycosuria is the secretion of glucose into the _____**Urine**
19. In mammalian kidney, the pyramids are seen in _____**Medulla**
20. Somatostatin _____ the secretion of insulin and glucagon _____**Inhibit**
21. The islets of Langerhans contain _____ **All of these (alpha, beta, delta and PP cells)**
22. Which of the following is a part of foregut_____? **Crop**
23. _____ of the blood volume entering kidney through renal arteries is removed as filtrate by glomeruli_____**10%**
24. Adrenal cortex synthesized and secret a family of steroid hormones collectively called____ _____**Corticosteroids**
25. An absolute deficiency of insulin due to loss of pancreatic beta-cell causes_____ _____**Diabetes mellitus Type-I**
26. ____ are involved in spermatogenesis_____**Germ Cells**
27. Insulin is secreted from the_____ of islets of Langerhans_____ **Beta cells**
28. Which of the following is an example of apocrine secretion? _____**Mammary glands**
29. Seminiferous tubules are located within the _____..._____**Testes**
30. Mammals have _____ adrenal glands_ _____**Two**
31. Salivary glands produce saliva that is delivered to the oral cavity through _____ _____**parotid and submandibular ducts**
32. An relative deficiency of insulin associated with defective insulin receptors causes -_____ _____**Diabetes mellitus Type-II**
33. Chromaffin cells are neuroendocrine cells found mostly in the _____ of the adrenal glands in mammals…_____**Medulla**
34. The type of the secretion in which entire cell rupture and breaks up to release its contents is called _____ **holocrine secretion**
35. The islets of Langerhans are found in_ _____**Pancreas**

36. Delta cells constitutes _____ of islets _____ **10%**
37. Major stimulus for the beta cells to secret insulin is high blood ____ level _____ **Glucose**
38. Adenohypophysis contains ….types of glandular cells that synthesize and secrete six hormones_____ **5**
39. The cell body of Neuron is also called _____._____ **Soma**
40. The Myelin sheath is derived from the _____._____ **Schwann cells**
41. are delicate threads running from cell body into the axon and dendrites._____ **Neurofibrils**
42. Goldman equation is an equation used to calculate the electrical equilibium potential across the cell's membrane in the presence of ions. _____ **More than one**
43. Cell body of neuron contains_____ **All of the above**
44. Delayed rectifies are ———-—-—-——— channels _____ **Voltage-gated K+**
45. Nissl's bodies consist of a group of _____ and _____ which are associated with protein Synthesis _____ **Ribosomes, Rough endoplasmic reticulum**
46. The__ is the region from which the axon of a neuron originates _____ **Axon Hillock**
47. Neuron has ….basic components _____ **Three**
48. Axons contain a jelly-like semi-fluid substance called _____ **Axoplasm**
49. Phases of action potentials include… _____ **All of the above**
50. _____ are Known as “delayed rectifiers” …_____ **Voltage-gated potassium channels**
51. _____are also called as spikes and nerve impulses _____ **Action potentials**
52. A. L. Hodgkin and A. F. Huxley carried out experiments on giant axons of squids in…_____ _____ **1940s and 1950s**
53. The period during which no new action potential can be fired is called the _____ _____ **absolute refractory period.**
54. Tetrodotoxin is a neurotoxin obtained from the Japanese…_____ **Puffer fish**
55. According to Nernst Equation, the equilibrium potential of a diffusible ion depends on. _____ **All of the above**
56. In many invertebrate species, action potentials can last for__milliseconds…_____ **10-100**
57. Most neurons have a resting membrane potential of _____ when no impulse is being conducted _____ **-70mV**
58. Cell body of neuron range in size between-_____ **5um – 135um**
59. Major ions responsible for Vrest are _____ **K^+, Na^+ & Cl**
60. _____ is the plasma membrane of an axon _____ **Axolemma**
61. Threshold potentials of most neurons fall between___and ____ _____ **-30 mV, -50 mV**
62. _____are the type of signals that neurons use to send information over long distances…_____ **Action potentials**
63. The permeability of Na+ channel for K+ ion is ------_____ **0.09**
64. Most neurons have a RMP of …. when no impulse is being conducted_____ **-70mV**
65. The receptors for umami taste were discovered in the year _____ **2000**
66. Internal stimuli include Stimuli of_____ _____ **Pain**
67. Major Neuromodulators in the Central Nervous System include_____ **All of the above**
68. Humans have more than _____ different types of receptors for bitter taste._____ **30**
69. Cornea is the clear outer surface of the _____ **eye**
70. _____ produce tears that is delivered through _____ on the surface of eye._____ _____ **Lacrimal glands, lacrimal duct**
71.

GRAND QUIZ/MID TERM

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35. Delta cells constitutes _____ of islets…_____ **10%**
36. Major stimulus for the beta cells to secret insulin is high blood ____ level _____ **Glucose**
37. _____is the process of converting that sensory signal to an electrical Signal _____ **Sensory transduction**
38. Membrane potential is expressed in…_____ **Millivolts**
39. Growth hormone is a protein hormone f about___ amino acids in a single chain _____ **191**
40. Cold Receptors are _____ times more common in skin than heat receptors _____ **3.5**
41. _____ secretes progesterone and estrogens…_____ **Corpus luteum**
42. Antidiuretic Hormone is also known as…_____ **Vasopressin**
43. Thickness of cell membrane is —-_____ **5nm**
44. _____ are cyclic unsaturated hydroxy fatty acids _____ **Prostaglandins**
45. In positive feedback systems the end product __the process of its production_____ **Speeds up**
46. An _____ is a simple, painless test that measures your heart's electrical activity…_____ **Electrocardiogram**
47. _____ is the phenomenon of unequal distribution of ions across the two sides of a differentially permeable membrane…_____ **Donnan Equilibrium**
48. Neurohypophysis secretes… _____ **Vasopressin. Oxytocin**
49. _____ are coupled transporters that transfer two solutes in opposite directions _____. **Antiporters**
50. Prostaglandins are synthesized in cellular membranes from ____ chains…__ **20-carbon fatty acid**
51. _____ are also called water channels protein…_____ **Aquaporins**

52. A membrane potential is the difference in electrical charge between…_____ _____ _____ _____ **The inside and outside of the cell**
53. Gonadotropin releasing hormone stimulates the release of…_____ **FSH and LH**
54. Olfactory receptors are located inside the…_____ **Nasal cavity**
55. The green, yellow. orange. and red colors lie between wavelengths of…_____ **525 and 675 nm**
56. Concentration of Potassium is ___times higher inside the cell than outside…_____**10-20**
57. Concentration of sodium is ___times higher inside the cell than outside…_____**20-10**
58. A goiter is an abnormal enlargement of your…_____**Thyroid gland**
59. Thermoreceptors are located in…_____.**Skin**
60. Melatonin is secreted by…_____**Pineal gland**
61. Adenohypophysis secretes _____tropic hormones…_____.**Four**
62. The Anterior lobe of pituitary gland is also called as…_____.**adenohypophysis**
63. Mechanisms through which carrier proteins actively transport substances are called…_____**Membrane pumps**
64. The G proteins family consists of about _____ different proteins that are composed of three subunits…_____**20**
65. _____short stature that results from insufficient secretion of Growth Hormone during childhood and adolescence…_____**Dwarfism**
66. Clotting of blood in response to an injury is an example of.…_____**Positive feedback system**
67. Mammary glands produce milk that is delivered through _____ to the nipples…_____ _____ _____**Lactiferous ducts**
68. Melatonin hormone synthesis_____ as the light _____ **increases, decreases**
69. Photopsins are the photoreceptor pigments found int he cone cells of the retina that are the basis of…_____ **color vision**
70. Radioactively labeled TTX is used to estimate density of ____ channels _____.**Na^+**
71. Naloxone is a …_____**Drug**
72. Saltatory Conduction Increases the velocity of nerve transmission_____ fold _____**5 to 50**
73. Protein and peptide hormones are synthesized on the _____ of endocrine cells…_____ _____ **rough endoplasmic reticulum**
74. External stimuli include Stimuli of…_____**Light**
75. Membrane potential values vary with type and physiological state of the cell, ranging between millivolts…_____**30 to -100**
76. _____is a thick, multi-layered membranous structure that surrounds many vertebrate axons…_____ **Myelin Sheath**
77. Visible spectrum for human eyes lies between the wavelengths of _____**400-740nm**
78. Pineal Gland are found on the dorsal surface of ____ in vertebrates _____ **forebrain**
79. Action potentials can travel as fast as _____ in large axons _____**120 m/s**
80. Action potential can last for _____ milliseconds _____**-100**
81. _____are the Junctions between axon terminals of one neuron and the dendrites of another neuron…_____ **Synapses**
82. _____is the movement of a substance from an area of high concentration to an area of low concentration…_____**Diffusion**
83. Facilitated diffusion is the _____ diffusion through the membrane down the concentration gradient _____._____**Passive, without**
84. Adrenocorticotropic Hormone is a peptide hormone comprising single chain of _____**39**
85. Exophthalmia is the abnormal protrusion of _____**Eyeball**
86. _____is an external or internal factor that can provoke response _____ **Stimulus**
87. Donnan Equilibrium phenomenon was discovered by Frederick Donnan in _____**1911**
88. The human ear can detect sound frequencies lying between hertz _____**20 to 20000**
89. _____ feedback system that end product works to stop or slow down that process…_____**Negative**
90. _____is also Known as hypophysis…_____ **pituitary gland**
91. _____is the contraction during which muscle length changes while muscle tension remain constant…_____**Isotonic Contraction**
92. _____is the hypersecretion of Growth Hormone before puberty which causes excessive size and stature…_____ **Gigantism**
93. The cell body of the neurons is also known as…_____ **Soma**
94. Which of the following is not an example of Amine hormones…_____**Cortisol**
95. _____ study of functions of tissue, organ and organ system of an individual_____**Physiology**
96. Glands are broadly classified into _____ types…_____**two**

97. _____ are Coupled transporters that transfer solutes in opposite directions…__ _____ **antiporters**

98. Symporters are coupled transporters that transfer two solutes in the_____direction _____ **same**.

99. _____ is the most familiar fast acting neurotransmitter_____ **Acetylcholine**

100. __of digestive tract is specialized for grinding and mixing _____ **Smooth muscles**

101. _____ helps in uterine contractions during child birth _____ **Oxytocin**.

102. _____is an immune system disorder that results in the overproduction of thyroid hormones… _____ **Graves' disease**

103. Which of the following gland is regarded as master a gland _____ **Pituitary gland**.

104. In humans most of ___is controlled by Adrenocorticotropic hormone _____ **skin pigmentation**

105. Human retina contains _____rods and 6 million cones _____ **125 million**

106. Neuron has _____ basic components _____ **Three**

107. Plasma membrane __ different concentration of ions inside and outside the cell ____ **Maintain**

108. skeletal muscle cells are _____ **multinucleate**.

109. Walter Cannon coined the term homeostasis in _____ **1929**

110. _____ is the study of physical laws that explain the relationship between pressure and flow of blood through blood vessels of the circulatory system_ _____ **Hemodynamics**

111. _____ is the hypersecretion of growth hormone after maturity which cause enlargement of the bones of the head and of the extremities _____ **Acromegaly**

112. Average cardiac output for resting adult human is _____ **5 L/min**

113. According to Nernst equation, the equilibrium potential of a diffuse able ion depends on _____ **Absolute temperature**

114. Which of the following is a correct statement_____ **Symporters are coupled transporters that transfer two solutes in the same direction.**

115. Adaptation is_____ _____**A slowly occurring evolutionary process**

116. In humans Melanocyte Stimulating Hormone is released from _____ of anterior pituitary in extremely low quantities _____ **Adenohypophysis**

117. _____ is a device that graphically records changes in the mechanical activities of animal tissues _____ **A kymograph**

118. Catecholamines hormones included _____**epinephrine and norepinephrine**

119. The receptors for umami (savory or delicious) taste were discovered in _____**2000**

120. ____ are projections of a neuron receive signals from other neurons_____ **Dendrites**

121. _____are synthesized from the amino acids tyrosine _____**Amine hormones**

122. Where the pituitary gland situated _____ **below the hypothalamus**

123. Control of blood glucose by insulin is an example of _____ **Negative feedback systems**

124. Humans have are more than _____ different receptors for bitter taste _____**30**

125. A person with loss of _____cones is called a protanope _____**Red**

126. Anterior lobe of pituitary gland contain _____ types of glandular cells that synthesized and _____ secret hormones _____ **five, six**

127. ___are specialized to pump and circulate blood throughout the body _____**Cardiac muscles**

128. H. E. Huxley and A. F. Huxley proposed the sliding filament theory of muscle contraction in _____**1954**

129. Which of the following hormone responsible for the “fight-or-flight” response _____ _____. **Epinephrine and Norepinephrine**

130. The posterior pituitary stores and release _____**Oxytocin and Antidiuretic Hormone**

131. Uterine cycle involves cyclic changes in the uterus. This cycle averages ___days _____**28**

132. Pituitary Gland is a small gland, about 1 centimeter in diameter and__ in Weight_____ ___ _____ _____**0.5 to 1 gram**

133. _____ are microscopic gaps found within myelinated axons _____**Nodes of Ranvier**

134. Endocrine glands are the ductless glands that secrete their products__ **directly into the blood**

135. ____ myelinated axon has faster speed than _____ thicker squid giant axon…____ **20pm, 40x**

136. Gastrin stimulates the _____ secretion in stomach _____**HCL**

137. _____ of testes produce and secrete the hormone inhibin _____**Sertoli cells**

138. Major stimulus for the beta cells to secrete insulin is high blood _____ level.___ Glucose

139. Which of the following hormones are responsible for the "fight-or-flight" response?____ _____Epinephrine and norepinephrine

140. _____ is also called somatotropin._____ Growth Hormone

141. A taste bud is composed of about _____ types of modified epithelial cells._____ 50

142. Gonadotropic hormones are produced by _____._____ **Adenohypophysis**

143. _____ is the process in which an individual organism adjusts to a change in its environment._____ **Acclimatization**
144. _____ are specialized for pumping and circulating blood._____ **Cardiac muscles**
145. Thyrotropes are endocrine cells in the anterior pituitary which produce _____. _____ **thyroid-stimulating hormone**
146. _____ is the tendency of an organism to maintain relative internal stability.__ **homeostasis**
147. Adrenal cortex synthesizes and secretes a family of steroid hormones collectively called _____. **Reproductive hormones**
148. Calcitonin is a hormone of which of following?_____ **Thyroid gland**
149. Testosterone is produced by___ _____._____ **Adrenal cortex**
150. Pituitary Gland is often called as “master gland” secreting _____ hormones._____ **Nine**
151. Cytosolic conc. of Ca2+ remains ----------._____ **Below 10-6 M**
152. If fertilization does not happen, corpus luteum degenerates after____ days in humans.____ **14**
153. __ occurs when thyroid gland produces too much of the hormone thyroxine._
154. A frog’s feeding process involves_____. **All of the above**
155. An opsin molecule has _____ transmembrane domains._____ **seven**
156. The hormones produced in the islets of Langerhans are _____. **insulin, glucagon**
157. _____ is the coiled sensory structure in the inner ear that is specialized for hearing._____ **Cochlea**
158. _____ is the Quantity of blood pumped into the aorta each minute by the heart._____ _____ **Cardiac Output**
159. That type of secretions in which entire cell ruptures and breaks up to release its contents is called as ----------. _____ **Holocrine Secretion**
160. Calcium level in the blood is increased by the_____. _____ **Parathyroid**
161. Thyroxine hormone is also known as _____. _____ **T4**
162. Antidiuretic hormone can increase water permeability of renal collecting duct in mammals up to _____times._____ **10**
163. __ is the movement of a substance from an area of high concentration to an area of low concentration._____ **Diffusion**
164. Continuous Capillaries are located in _____. **All of the above**
165. Aldosterone is produced in the _____ of the adrenal glands._____ **cortex**
166. …….. is the study of functions of tissues organs systems of animals _____ **Physiology**
167. The amount of solute that passes through a unit area of membrane every … in one direction_ _____ **Second**

FINAL TERM

QUIZ NO 2

1. Which of the following is an example of zymogen? _____ **Chymotrypsinogen**
2. Which of the following is a part of foregut?_____ **crop**
3. Which of the following statement is correct for esophagus?___ **A sac-like expanded section**
4. ……. hormone is also referred as hunger hormone?_____ **Ghrelin**
5. …….. functions best at a low pH value of 1-2?_____ **Pepsin**
6. Which of the following parts of Digastric Stomachs act as fermentation chambers? _____ **rumen and reticulum**
7. In Food handling and Mastication …….. are involved. _____ **Teeth and tongue**
8. ……. Is involved in process of defecation._____ **Hindgut**
9. The intestinal folds, villi and microvilli collectively increase the surface area more than. **500 times**
10. Which of the following is an endocrine secretion of GIT?__ **Gastric inhibitory peptide (GIP)**
11. Digastric stomach are found in?_____ **Artiodactyla**
12. ……. concentrates and stores bile._____ **Gallbladder**
13. human small intestine is …… long. _____ **6 meters**

14. Bile assists in emulsification of._____ **Fats**
15. Mechanism of Ciliary motility is used by._____ **cephalo-chordates**
16. Once food reached back of buccal cavity swallowing becomes involuntary under direct neural control of _____ **medulla oblongata**
17. Initial stages of protein digestion are carried out at_____ **Stomach**
18. Hemoglobin is called oxyhemoglobin when ____ is bound_____ **Oxygen**
19. The average total lung capacity of holding of air in an adult human male is about_____**6000 mL**
20. The average total lung capacity of holding of air in an adult human female is about____**4200 mL**
21. Alveoli are tiny air sacs in the_____**Lungs**
22. Most diving animals have _____ levels…_____**High hemoglobin and myoglobin**
23. _____ are located in the medulla of mammals and many other air breathing vertebrates _____**Central Chemoreceptors**
24. Osmolytes are substances which reduce the need to maintain osmotic pressure with ___ ions _____ **Inorganic**
25. Pulmonary surfactant is a surface-active lipoprotein complex formed by _____Alveolar cells _____**Type II**
26. _____ is the active regulation of the osmotic pressure of an organism's body fluids… _____**Osmoregulation**
27. Lung volumes and lung capacities refer to the volume of air in the lungs at different phases of the_____**Respiratory cycle**
28. Thromboxane A is synthesized by _____ and causes vasoconstriction_____**Platelets**
29. Which one of the followings is not a respiratory pigment?. _____ **Charcoal**
30. A condition in which the body or a region of the body is deprived of adequate oxygen supply is called _____**Hypoxic**
31. Blood with hemoglobin can carry 20 ml oxygen per 100 ml at _____ sea level _____**37°C**
32. Air capillaries are the sites where oxygen and carbon dioxide are _____._____**Exchanged**
33. Do not possess a diaphragm… _____ **Reptiles**
34. Peripheral chemoreceptors are located in… _____ **All of the above**
35. _____ oxygen binds with hemoglobin… _____**97%**
36. Angiotensin II is produced in… _____**lungs**
37. Lung ventilation is a mechanical process and it consists of _____ phases _____ **Two**
38. A male human can forcefully expel additional ___ air by use of accessory muscles…__.**1000 mL**
39. A female human can forcefully expel additional ___ air by use of accessory muscles… __**700 mL**
40. Lungs are located in…_____**Thoracic cavity**
41. Gas transfer system involves _____ basic steps…_____**Four**
42. Alveoli are very small…_____**bubble-like**
43. _____ is the phase of ventilation in which air enters into the lungs…… **Inspiration**
44. If all sites on hemoglobin molecule are occupied by Oxygen, blood is _ saturated….____**100%**
45. Amount of air that moves into or out of lungs during a single, quiet respiratory cycle is called _____**Resting tidal volume**
46. Fishes live in aquatic environment that contains less than _____ of oxygen than present in air _____**2.5%**
47. Respiratory surface area of human lung is _____**50 to 100 m2**
48. Affinity of hemoglobin for CO _____ times greater than for oxygen _____.**200**
49. Pulmonary Surfactants have a half-life of about ____ hours in mammals… _____**12**
50. Respiratory system of birds consists of _____ air sacs… _____**7 or 9**
51. The bronchioles are an important part of the _____**Respiratory system**
52. During Inspiration. the diaphragm _____**Contracts**
53. In expiration phase of ventilation air with low _____ and high _____ content is moved out… _____ **Oxygen, carbon dioxide**
54. Many air-breathing aquatic vertebrates go under water for varying periods of time is Termed as_____ **Dive**
55. Carbon dioxide transportation in blood occurs in ____ forms _____**Three**
56. _____ is a potent vasoconstrictor hormone…_____**Endothelin**

57. Fishes have gills which are efficient respiratory organs to extract ___ in water _____ **Oxygen**
58. Blood is _____times viscous than water._____ **3 to 4**
59. On the basis of their wall structure, capillaries are classified into _____types._____ **Three**
60. A _____is a complex of arteries and veins lying very close to each other, found in some vertebrates._____ **Rete mirabile**
61. _____ is the Volume of the blood ejected by each beat of heart"_____ **Stroke Volume**
62. Fenestrated capillaries are found in _____.**Renal glomerulus, Intestines, Endocrine glands**
63. According to _____flow of blood is inversely related to viscosity._____ **Poiseuille's law**
64. What does vasoconstriction stand for?_____ **Narrowing of the blood vessels**
65. Which type of blood vessels carry blood away from the heart?_____ **Arteries**
66. Which one of the following is not associated with classification of capillaries?_____ _____ **Exocrine capillaries**
67. Blood pressure is generated by the _____of ventricle of heart._____ **Contraction**
68. The animal's circulatory system alternatively can be called as_____ ._____**The body's blood-transporting system**
69. Capillaries are microscopic vessels which are about 1 mm long and ___in diameter.____ **3-10 µm**
70. Large number of rete mirabile in tuna (fish) are used to regulate temperature of _____.**Brain**
71. Phenomenon of plasma skimming and resultant increased blood flow velocity is called _____ _____.**Fahraeus-Lindqvist effect**
72. _____serves as a major inducer of vasodilation in cardiovascular system._____ **Nitric oxide (NO)**
73. In Peripheral circulatory pathway blood is pumped from _____.**Left ventricle**
74. After death, human and other animal's bodies gradually become rigid. This condition is called _____**rigor mortis**
75. _____is the cytoplasm of a muscle fibre…_____ **Sarcoplasm**
76. The organs of vertebrates which regulate water and ions are_____.All of the above
77. H. E. Huxley and A. F. Huxley proposed Sliding Filament Theory in _____ to explain muscle contraction._____ 1954
78. Which is most likely to extend the entire length of a muscle fibre?...._____**Myofibril**
79. Hypercapnia is a condition of abnormally elevated _____levels in the blood.____ **Carbon dioxide**
80. Peripheral chemoreceptors are located in_____ _____.**All of the above**
81. Fishes have gills that are efficient respiratory organs to extract ___from water._____ **Oxygen**
82.

Quiz no 3

1. Hepatocytes are the cells of_____ **Liver**
2. Birds maintain a core body temperature nearly…_____**400C**
3. Amylases are secreted by _____ and pancreas _____**Salivary glands**
4. _____hormone is also referred as "hunger hormone" _____ **Ghrelin**
5. Proteases are enzymes that break the peptide bonds of_____**Proteins**
6. In mammals gastric secretion occurs in____phases… _____**Three**
7. Warm blooded animals maintain a core body temperature_____between _____**37-38°C**
8. _____lack high rates of heat production _____**Poikilotherms**
9. Which of the following is an example of zymogen?..._____**Chymotrypsinogen**
10. _____attack terminal peptide bonds of shorter polypeptides _____**Endopeptidases**
11. Gastric juice contains _____**All of the above**
12. 12. _____Functions best at a low pH value of 1-2 _____**Pepsin**
13. 13. Organic salts of bile are manufactured in___ from cholesterol and amino acids complexed with sodium… _____**Liver**
14. 14. In vertebrates ____ chylomicrons enter lacteals of villi _____**80%**
15. 15. Bile salts are essential for the digestion of _____._____**Fats**
16. 16. Which of the following is an endocrine Secretion of GIT?.._____ _____ _____ _____**gastric inhibitory peptide (GIP)**

17. 17. A lizard Needs _____ lesser food per day than a rodent of Same size living in the same habitat and having the same diet _____ **17 times**
18. 18. _____ attack peptide bonds within protein molecule…_____ **Endopeptidases**
19. 19. _____ is a gastric proteolytic enzyme produced by chief cells of stomach released in its inactive form called _____ **Pepsin. pepsinogen**
20. 20. Bile is transported by _____ from liver to the gallbladder.… _____ **Hepatic duct**
21. _____ are the animals that generate their own body heat through metabolism to elevate their body temperature…_____ **Endotherms**
22. 22. _____ are also called warm blooded animals _____ **Homeotherms**
23. 23. Brunner's glands are present in the first part of _____ between pyloric sphincter and pancreatic duct _____ **Duodenum**
24. Non diffusible ___produce Donnan effect_____ **SOLUTE**
25. . ____ channels are involved in producing action potential._____ **Na+ and K+**
26. AChE abundantly present in _____ **SNYPTIC CLEFT**
27. Which ___monitors the position of muscles and joints._____ **PROPRIOCEPTORS**
28. Sensory sensilla are organ of ___in insects._____ **TASTE**
29. Merkel Disks are associated with the reception of _____ **VIBRATION**
30. Human ear can detect sound _____ **20 TO 20000 HRTZ**
31. interstial tissues constitute___ of adult testes mass_____ **20%**
32. Velocity of impulse conducting by arterial fiber is _____ **0.3 m/s**
33. Equation indicate blood flow _____ **V = Q/A**
34. Relative viscosity of plasma _____ **1.8**
35. Fenestrated capillaries are found in _____ **endocrine glands**
36. Which is local vasoconstrictor _____ **Angiotensin II**
37. ___ is formed in lungs, acts as local vasoconstrictor _____ **Angiotensin II**
38. Amount of air M/V that still remains in lungs when they fully collapse _____ **30 to 120mL**
39. Respiratory system of ___ is composed of 7 or 9 air sacs_____ **Birds**
40. Central chemoreceptors are located in ____Medulla of mammals and Air breathing __ **vertebrates**
41. which are stenohaline animals___ _____ **Marine spider, crab**
42. A camel can lose ___ of water and still survive _____ **40%**
43. ___ is involve in maintaining pH of urine_____ **TUBULAR SECRATION**
44. GFR averages an astounding ___ per minute_____ **125 mL**
45. It requires only __ of water to excrete 1g of urea._____ **0.05 L**
46. Esophagus is included in ___ division of mammalian digestive tract._____ **Four (4)**
47. Human GIT secrets__ exocrine fluid daily._____ **Bile and Bile salts**
48. Goblet cells synthesis and secretes_____ **Mucus**
49. ___ is largest water soluble essential nutrient _____ **Vitamin B12**
50. Brown fat contains large amount of _____ **Mitochondria**
51. Which is Neuroethological behavior_____ **Study behavior in laboratory**
52. Agonistic behavior involve___b/w animals_____ **Competition**
53. Social behaviors are based on _____ **Competition**
54. Respiratory epithelium has only a thickness of _____ **0.5 to 15 um**
55. High pressure is _____ **Systolic BP**
56. Atrial natriuretic peptide (ANP) hormone secreted by _____ **Stretch-Sensitive Secretory Cells**
57. Blood with hemoglobin carry ___ Oxygen per 100ml._____ **20 ml**
58. Inspiratory volume is _____ **1900 ml**
59. DCT secrete _____ **NH4+ ions, K+ ions, & H+ ions**.
60. Teloest fish is _____ **Ammonotelic Animals**
61. carrier protein Na+/K+ _____ **ATPase**
62. Giant axons of arthropods and molluscs: _____ **30 m/sec**
63. Which of the following is not a pharmacological application _____ **Anemia**
64. Calcium ions are not _____ **delayed rectifiers**
65. Example of mechanical stimuli is _____ **gravity**
66. Which one is not G protein receptor for taste _____ **sour and salty**
67. Merkel's Disks are associated with the reception of _____ **vibration**
68. Meissner's Corpuscles receptors are for _____ **touch**
69. Each hair cell has 20-300_____ **nonmotilestereocilia**.
70. Human ear can detect sound frequencies lying between _____ **20 to 20000 hertz.**
71. Hair cells have a Vrest of _____ **-60 mV**
72. Sensory layer of eye with photoreceptor cells Rods and cons is_____ **Retina**

73. Visible spectrum for human eyes lies between the wavelengths of _____ **400-740 nm**
74. A person with loss of red cones is called a _____ **protanope**
75. Effect on neighboring cells_____ **paracrine secretion**
76. Adenohypophysis has pituitary lobe _____ **anterior**
77. ACTH is a peptide hormone comprising single chain of _____ **39 amino acids**
78. Hypothyroidisim causes _____ **Cretinism and Goiter**
79. Regulate Ca2+ and phosphate ions in blood _____ **Parathormone PTH**
80. CatecholaminesConversion of tyrosine to dopa and dopamine occurs in the_____ **cytosol**
81. Thyroglobulin is stored within the thyroid gland in large _____ **follicles**
82. Diameter of membrane channel is less than _____ **1.0nm**
83. channel involve producing action potential _____ **Na+ and K+**
84. Action potential is inflow of Na+ _____ **depolarization**
85. Hallucination _____ **Endorphins**
86. The receptor molecules are typically_____ **proteins**
87. sensory sensilla are organ of_____ **taste**
88. Prolactin regulates fat metabolism and reproduction in _____ **Birds**
89. Example of peptide hormone in reproduction hormones _____ **Oxytocin**
90. Rod photoreceptors are sensitive to _____ **low intensity dimmer light**
91. The correct statement is Symporters are coupled transporters that transfer two solutes in the same _____ **direction**
92. Voltage-Gated Na+ Channels are _____ **Fast Acting channels**
93. Delayed Rectifiers are _____ **Voltage gated K+ channels**
94. Cytosolic conc. of Ca2+ remains below _____ **10-6M**
95. Thickness of cell membrane is _____ **5nm**
96. Which is not associated with ATPase_____ **Ca Activated**
97. Which is not Fast inhibitory transmitters _____ **Acetylcholine**
98. AChE is abundantly present in the _____ **synaptic cleft**
99. Example of steroid hormone is _____ **Aldosterone**
100. substance that is released via a duct through the gland to the external or internal epithelial surfaces_____ **Exocrine secretion**
101. Inhibits release of Growth Hormone (GH) _____ **Somatostatin**
102. Amine hormone (Thyroid & Adrenal Modulary hormone) are synthesized by the action of Enzyme on amino acid_____ **Tyrosine**
103. Acts as enteric neurotransmitter_____ **Substance P**
104. In many organisms, ACTH also plays role in_____ **circadian rhythms**
105. Pituitary Gland release __ hormones_____ **9**
106. Which is not part of a cell body _____ **Myelin Sheath**
107. Adenohypophysis It contains ___**five**___ types of glandular cells that synthesize and secrete **six** ____hormones
108. A color-blind person who lacks ___ ____ cones is called a deuteranope_____ **green**
109. Example of Apocrine secretions _____ **Mammary gland**
110. Salivary glands produce saliva that is delivered to the oral cavity through _____ _____ **submandibular and parotid duct**
111. Oxytocin stimulates motility of the oviduct in _____ **Birds**
112. Which of the following hormone is released from anterior lobe of pituitary gland__ luteinizing hormone _____ **LH**
113. Parathormone over secretion leads to _____ **Softness of bones**
114. Type of secretion in which entire cell raptures and breakup to release its content_ _____ _____ _____ **Holocrinesecration**
115. Human retina contains ___ ___ cones_____ **6 million**
116. Which one is not related to diffusion option_____ **low to high concentration**
117. melatonin belong to_____ **pineal Gland**
118. pineal gland found dorsal surface of.....in vertebrate _____ **forebrain**
119. target cell of hormones usually have ....receptors _____ **2000 to 100,000**
120. Thick filament are composed of ...._____ **myosin**.
121. Each molecule of calsequestrin protein can find around.....calcium_____ **50**
122. human s-A node is ...thick_____ **1 millimeter thick**
123. Velocity of impulse conduction through most atria fiber is.....____ **0.3 m/s.**
124. Myocytes are ____ muscles _____ **Cardiac**
125. The knob of dendrite has ____olfactory cilia_____ **4 to 25**
126. Synaptic cleft is ___ nm wide _____ **20**

127. Pituitary gland is __ dm _____ **1cm**
128. Rhythmical discharge of S-A node is: ____/mint _____ **70 to 80**
129. An adaptive change within an individual animal. Happens due to chronic exposure to new, naturally occurring environmental conditions_____ **Acclimatization**.
130. Antidiuretic hormone can increase water permeability of renal collecting duct in mammals up to _____ **10 times**
131. _____ are delayed rectifiers _____ **Voltage – gated K+channels**
132. Which one of the following is not the feature of the Voltage –gated Ca+channel _____ _____ ___ **They are known as delayed rectifier**
133. Which of the following statement is correct about action potential _____ _____ **inflow of sodium ion (depolarization)**
134. Which of the following monitor the position of muscles and joints_____ **Proprioceptors**
135. Taste receptors in terrestrial vertebrates are not located in/on _____ **Larynx**
136. Which is not the part of organ of Corti_____ **Reissner's men**
137. Near the end of pregnancy, placental estrogens are 30 times more than the normal levels. _____ **30 times**
138. Myofibers are ___ in diameter, and up to many centimeters in length_____ **5 to 100 µm**
139. Troponin has high affinity for_____ **Ca2+.**
140. When the force exerted by the muscle contraction is equal to the opposing external force than ___ contraction _____**isometric**
141. Atrial contraction provides only __volume of total mammalian ventricular output_____**30%**

MID TERM

1. What is the basic theme in physiology 2? marks

Animal Physiology deals with the study of functions of the tissues, organs and organ systems of animals.

Our major goals of study of animal physiology are to:
explore the physiological processes that are basic to all animal groups show how they have been shaped by selective forces during evolution

Central themes in animal physiology:
As we study animal physiology and physiological adaptations, we notice several basic themes, repeatedly emerging.

We shall briefly discuss five major themes here, i.e.

(I) Structure-function relationships
(ii) Adaptation, Acclimatization and Acclimation
(iii) Principle of homeostasis
(iv) Feedback control systems
(v) Conformity and Regulation

2. What is the disease caused by hyperthyroidism.2marks?

Excessive secretion of thyroid hormone is known as hyperthyroidism.
The most common form of hyperthyroidism is **Graves'** disease. It is an autoimmune disorder in which antibodies that mimic TSH bind to the receptor for TSH and cause sustained thyroxine production.

- high body temp.
- profuse sweating
- weight loss
- Irritability
- high blood pressure
- protruding eyes (exophthalmia)

3. What is the function of fibers of zonula and ciliary muscels.3marks?

In higher vertebrates, image is focused by changing the curvature and thickness of the lens. The curvature and thickness of the lens is controlled by the suspensory ligaments (fibers of Zonula) and ciliary muscles.

Role of Fibers of Zonula

The shape of the lens is changed by the fibers of the zonula that held the lens in place.

These fibers can exert an outwardly directed tension on the perimeter of the lens.

Role of Ciliary muscles

Ciliary muscles are attached with the fibers of zonula.

Their contraction and relaxation adjust the amount of tension exerted on the lens.

When the ciliary muscles contract, the lens is flattened by elastic tension exerted by the fibers of the zonula, which pull the perimeter of the lens outward. In this state, objects far from the eye are focused on the retina, but objects close by would be fuzzy. When the ciliary muscles relax, lens becomes more rounded. This focuses the objects close to the eye.

4. Write about the genetic basis of color blindness.3marks

Genetic Basis of Color Blindness

The three types of opsins found in color pigments are encoded by three different genes.
The gene encoding the opsin in blue-absorbing pigment is located on an autosomal chromosome. The genes for red-absorbing and green-absorbing pigments are closely linked on the X chromosome. Color blindness is caused due to a mutation in one of the cone opsin genes, resulting in absence of one type of pigmented cones.
A person missing a single type of color receptive cones is unable to distinguish some colors.

5. What is meant by range fractionation, also give its importance and give at least two examples.5marks

Definition

In a sense organ, sensory receptors are arranged in an order of increasing sensitivity to different range of intensities of the stimulus. This hierarchical arrangement of receptors is known as range fractionation.

Range Fractionation and Stimulus Intensities

Each individual receptor, in a sense organ, covers only a fraction of the total dynamic range of the stimulus. Receptors work together in an orderly way to provide discrimination of stimulus intensities.

Recruitment Phenomenon

One important implication of range fractionation is the recruitment phenomenon.

• The most sensitive receptors in the population produce a response at stimulus intensities that are just above the threshold. Above that intensity, the most sensitive receptors become saturated.

- If the stimulus energy is increased a little, the less sensitive receptors in the population will join in and respond.
- With still greater stimulus intensities, another, formerly inactive lower-sensitivity population of receptors will join in.
- In this way, as the stimulus intensity is increased, receptors that are less and less sensitive will become active, until the least sensitive sensory fibers will finally be recruited, and all receptors will respond maximally.
- At that point, the system will be saturated and therefore unable to detect further increases in intensity. This phenomenon of activation of receptors in a graded fashion is called recruitment.

Importance of Range Fractionation

- It results in increasing the active range of multineural sensory system than the range of any individual single receptor.
- It also increases the overall precision of the sense organ.
- It enables the sensory processing centers of the CNS to discriminate stimulus intensities over a range much greater than that of any single sensory receptor.
- The extended dynamic range of the entire system is possible because individual receptors of a sensory system cover different parts of the full spectrum of sensitivity.

Range Fractionation Example:

Eyes have rod cells which are high sensitivity photoreceptors while cone cells have low sensitivity. At low light intensity, i.e. dimmer light, only rod cells are recruited and we can see dimly, as black/white image. As the brightness increases, cone cells start to recruit and we are able to see colors.
Very bright light gives us perception of truly colored image.
After the highest intensity limit, increasing intensity does not increase clarity of the image because the system becomes saturated.

6. Write at least five properties of hormones.5 marks

1.Hormones act only on specific target cells/tissues
2. hormone molecules come into contact with all the tissues in the body during transport through blood, only the cells that contain receptors specific for a particular hormone are affected by it, and generate response.
3.Hormonal action depend on the type of receptor

4.The action of a hormone depends on the nature of enzyme cascade linked to the hormone's receptor, as well as the effector molecules expressed in a particular tissue.

For this reason, a hormone can act on two or more different types of tissues and generate different types of responses.
For example, Antidiuretic hormone stimulates water reabsorption in the nephron of kidney while the same hormone produces vasoconstriction and increases blood pressure by acting on the walls of arteries. The receptors in kidney nephron tubule and arterial receptors are different, causing the same hormone to generate different responses.

5. Hormones are active at very low concentration

The amount of hormone produced by an endocrine gland is small. It is further diluted in the blood and interstitial fluid. The available concentration for the target cell lies between 10-8 to 10-12 M and the hormone is still effective at this very low concentration.
The high sensitivity of hormonal signaling is due to the high affinity of target cell receptors for hormones.

6. Hormonal effects are amplified

Binding of a hormone molecule to its receptor leads to a cascade of enzymatic steps that amplify the effect; thus just a few hormone molecules can influence thousands or millions of molecular reactions within a cell.

7.Write note on Acetylcholine 5 marks

Acetylcholine

- The most familiar fast acting neurotransmitter substance is the Acetylcholine.
- In most instances, acetylcholine has an excitatory effect.
- It also acts as inhibitory transmitter in some instances e.g. at the peripheral parasympathetic nerve endings where it is involved in the inhibition of the heart by the vagus nerves.

Cholinergic Neurons

Neurons that release Acetylcholine are said to be cholinergic. They are widely distributed throughout the animal kingdom. These neurons include:

- vertebrate motor neurons
- preganglionic neurons of the vertebrate autonomic nervous system
- postganglionic neurons of the parasympathetic division of the autonomic nervous system
- many neurons of the vertebrate central nervous system
- a number of invertebrate neurons e.g. cells of the molluscan central nervous system, motor neurons of annelid worms, and sensory neurons of arthropods.
- **Mode of Action of Acetylcholine**

• When acetylcholine (ACh) is released into the synaptic cleft, it binds to the ligand-gated ACh-specific receptors in the postsynaptic membrane. • The binding causes Na+ and K+ ion channels to open briefly that produce an EPSP.

8.Photoreceptors 3 marks

Photoreceptors:

Photoreceptors possess light-sensitive pigments which are carotenoids retinal and 3-dehydroretinal associated with opsin proteins to form rhodopsins.

These pigments absorb photons of light energy and then produce a generator potential.

External energy, such as light, may strike any part of the body; but only the eyes contain sensory cells that can receive and respond to the stimulus of light and transduce photons into neuronal energy.

9.Basic type of stimuli.2marks

types of Stimuli

Two types:

1. **External stimuli** :Stimuli of odor, touch, light, sound and gravitation
2. **Internal stimuli** : Stimuli of pain, homeostatic imbalances and blood pressure

10.Stigma and ocelli 5 marks

Eyespot or Stigma

The simplest photoreceptive structures are the Eyespots or Stigma found in some protozoa e.g. euglena. Stigma is a bright red colored structure having carotenoid pigments. It gives a sense of light and dark and helps in phototaxis.

Eyecups or Ocelli

Cnidarians and flatworms have multicellular photoreceptive structures call eyecups or Ocelli. They consist of a cuplike depression containing photoreceptor cells. They cannot form image and only provide the animal a sense of direction.

11.Deuteronatope and paranatope 3 marks.

Protanope and Deuteranope

A person with loss of red cones is called a protanope.

In this type, the overall visual spectrum is shortened at the long wavelength end because of a lack of the red cones.

A color-blind person who lacks green cones is called a deuteranope.

Such a person has a normal visual spectral range because red cones are available to detect the long wavelength red color.

12.Donnan equilibrium 3marks

Discovery

In 1911 Frederick Donnan discovered the phenomenon of unequal distribution of ions across the two sides of a differentially permeable membrane. This phenomenon is known as Donnan equilibrium and can be defined as:

Definition "If diffusible solutes are separated by a membrane that is freely permeable to water and electrolytes but totally impermeable to one species of ion, the diffusible solutes become unequally distributed between the two compartments".

13.Role of corpus luteum in pregnancy.

Role of Corpus Luteum in Pregnancy

If the released ovum is fertilized and becomes implanted in the endometrium, the developing placenta begins to produce chorionic gonadotropin hormone that maintains an active corpus luteum, so that estrogen and progesterone secretion continues until the placenta fully takes over the production of these hormones, at which time the corpus luteum degenerates.

In many other mammals, such as the rat, the corpus luteum, stimulated by prolactin, continues to grow and secrete estrogen and progesterone throughout the gestation period.

14.Explain negative feedback with 2 example.5

The secretory activities of most endocrine tissues, especially those involved in maintaining homeostasis, are modulated by negative feedback.

In this type of feedback, the hormone itself or products resulting from its action tend to suppress its further release.

Negative feedback mechanisms ensure a proper level of hormone activity at the target tissue and prevent its overactivity by over-secretion.

Negative feedback is a type of regulation in biological systems in which the end product of a process in turn reduces the stimulus of that same process. Feedback, in general, is a regulatory mechanism present in many biological reactions. By allowing certain pathways to be turned off and on, the body can control various aspects of its internal environment. This is similar to flipping a switch. Feedback allows the product of a pathway to control the switch. Sometimes referred to as a "negative feedback loop", negative feedback occurs when the product of a pathway turns the biochemical pathway off

Negative feedback systems:

In life, the most common form of regulation encountered is the negative feedback, in which accumulation of an end product works to stop or slow down that process.

Example-1

The breakdown of sugar in the cells generates chemical energy in the form of ATP. When a cell makes more ATP than it can use, the excess ATP "feeds back" and inhibits an enzyme near the beginning of the pathway. This results in temporary stoppage of ATP production.

The figure illustrates the negative feedback:

The three-step chemical pathway shown here converts substance A to substance D. A specific enzyme catalyzes each chemical reaction. Accumulation of the final product (D) inhibits the first enzyme in the sequence, thus slowing down production of more D.

Example-2

The control of blood sugar (glucose) by insulin is another good example of a negative feedback mechanism.

When blood sugar rises, receptors in the body sense a change. In turn, the control center (pancreas) secretes insulin into the blood effectively lowering blood sugar levels. Once blood sugar levels reach homeostasis, the pancreas stops releasing insulin.

15.Abnormaities due to growth hormones.3marks

Developmental Abnormalities Due to GH

Disturbances in the secretion of growth hormone lead to several patterns of abnormal growth and development in humans:

Gigantism: excessive size and stature caused by hypersecretion of growth hormone during childhood (before puberty).

Acromegaly: enlargement of the bones of the head and of the extremities caused by hypersecretion of growth hormone beginning after maturity.

Dwarfism: abnormal underdevelopment of the body caused by insufficient secretion of growth hormone during childhood and adolescence.

16.Androgen hormone secreted by adrenal cortex.2marks

Hormones of Adrenal Cortex

The adrenal cortex produces two major types of steroid hormones: the mineralocorticoids and the glucocorticoids collectively called as corticosteroids. These hormones are involved in blood ion and glucose regulation and antiinflammatory reactions.

- Adrenal cortex also secretes small amounts of sex hormones, especially androgenic hormones.
- These androgens exhibit about the same effects in the body as the male sex hormone testosterone.

17.Give sliding filament theory, 5 marks

Sliding Filament Theory

H. E. Huxley and A. F. Huxley proposed the sliding filament theory of muscle contraction in 1954.

This theory states that during muscle contraction the thin and thick filaments in sarcomeres slide and undergo shifting.

When a muscle contracts, the thin actin filaments actively slide along between the thick myosin filaments and move closer to the center of the sarcomere. As a result, the sarcomere becomes shorter.

When a muscle relaxes or is stretched, the overlap between thin and thick filaments is reduced, and the sarcomere elongates.

The changes in sarcomere length during contraction and stretch of a muscle, correspond to changes in muscle length.

Explanation • In a relaxed muscle fiber, the thick and thin filaments overlap only at the ends of A band. But when muscle fibers are stimulated by the nervous system, myosin heads are attached to the myosin-binding sites on actin in the thin filaments, i.e. cross bridges are formed and the sliding begins.

head

- During contraction, the A bands (myosin filaments) maintain a constant length, whereas the I bands and the H zone (zones where actin and myosin filaments do not overlap) become shorter and Z lines get closer. • When the muscle is stretched, the A band again maintains a constant length, but the I bands and H zone become longer.
- Neither the myosin thick filaments nor the actin thin filaments change their lengths when a sarcomere shortens or is stretched. It is the extent of overlap between actin and myosin filaments that changes.

Length-Tension Curve

One of the strongest pieces of evidence in support of the sliding filament theory comes from the length-tension relation of a sarcomere.

Experimental measurement of the shortening of length of sarcomere during contraction and resulting force generates a length-tension curve. This curve explains the assumptions of sliding-filament theory.

Explanation of the Curve

- The tension produced by the muscle is maximal when the overlap between thick and thin filaments allows the largest number of cross-bridges to be formed between actin and myosin.
- Tension drops off with increased length of sarcomere, because the thick and thin filaments overlap less and fewer crossbridges can be formed.
- It also drops off with decreased length, because thin filaments begin to collide with one another, preventing further shortening. • The curve also predicts the consequence of sliding filament theory that no active tension will develop if a sarcomere is stretched so far that there remains no overlap between actin and myosin filaments, making it impossible to develop any crossbidges.
- **Conclusion**

This curve shows that the tension produced by a sarcomere is proportional to its shortening which is due to sliding of thick and thin filaments and formation of cross bridges in the sarcomere during contraction. These were the proposals of sliding filament theory. So length-tension curve provides a practical proof of this theory.

18.What are smooth muscle and there general function?.

Smooth Muscles General Features

- Least specialized muscle fibers.
- Have myosin similar to that found in contractile nonmuscle cells.
- Non-striated.
- Involuntary: under autonomic control.
- Contract and relax slowly.
- Capable of more sustained contractions.

19.Teste buds and types of cell teste buds.

Taste buds

- The gustatory organs of vertebrates are called taste buds.

The taste bud is composed of about 50 modified epithelial cells including supporting cells (sustentacular cells), basal cells and taste receptor cells.

- The basal cells are progenitor cells that give rise to new taste receptors. They regularly generate new sensory taste receptor cells which have an active life of only 10 days.
- **Taste Receptor Cells**
- The outer tips of the taste cells are arranged around a minute taste pore.
- From the tip of each taste cell, several microvilli, or taste hairs, protrude outward into the taste pore to approach the cavity of the mouth.
- These microvilli provide the receptor surface for taste.
- Interwoven around the bodies of the taste cells is a branching terminal network of taste nerve fibers that are stimulated by the taste receptor cells.

20. Sensory adaptation with example.

Definition

The decrease in perceived intensity, when the intensity of the stimulus has not itself changed, is known as sensory adaptation. The phenomenon is due to reduction in the frequency of sensory response during a sustained stimulus.

Adaptation Example:

Wearing clothing stimulates touch receptors at all points where our garments touch the skin. But we typically adapt to the touch input from our garments. So we can easily detect any new touch stimuli that impinge on our skin, even at locations covered by our clothing.

Other common examples of adaptation under our everyday observation are with the stimuli of smell, hot or cold water and dim or bright intensity of light.

More Examples

- Stimuli of smell
- Hot or cold water
- Dim or bright light

21.Feedback mechanism principle and significance.

Feedback Control Systems:

Definition:

Many biological processes have the ability of self-regulation by a mechanism which is called feedback mechanism. The basic principle of feedback in living systems is that the output or product of a process itself regulates the process.

Significance:

The feedback regulatory processes maintain homeostasis in the cells and the body of multicellular organism as a whole.

Mechanism: Feedback controls regulate particular variables e.g.,

- Temperature
- Salinity
- pH

This regulation requires:

- Continuous sampling of controlled variables
- Respective corrective actions.

22.What is homeostasis. 2marks

Homeostasis—definition:

During evolution, each species has assumed a specific set of internal environment with an ability to resist environmental changes by making adjustments to keep its internal fluctuations in a narrow range. This ability to protect internal environment from the harms of fluctuations in external environment is termed as homeostasis.

Homeostasis does not mean to keep a fixed internal environment as the changes maintained within a specific range are necessary for normal body functions.

Example:

1. Homeostatic regulation of water
2. Homeostatic regulation of temperature
3. Regulation of pH
4. Regulation of glucose concentration
5. Regulation of osmotic pressure 6. Regulation of oxygen level
7. Regulation of ion conc.

23.Phasic or tonic function .

Adaptation: "The reduction in the frequency of action potentials that is typically seen in a neuron during asustained stimulus". Two types:

- Phasic response
- Tonic response

Adaptation: Phasic response: The type of adaptation in which neurons adapt rapidly & stop generating more APs after initial 1 or 2 APs.

Adaptation: Tonic Response: The adaptation in which neurons adapt slowly & fire repetitively to a subthreshold stimulus, but with gradually decreasing frequency.

24.Calcium ion role in excitation.

Transmission of AP through T tubules results in the release of Ca2+ ions form stores of sarcoplasmic reticulum. This increases Ca2+ conc. inside the muscle fiber in the immediate vicinity of myofibrils.

- These calcium ions cause the contraction to begin. So, the net effect of excitation-contraction coupling is to link an AP in the plasma membrane of the muscle fiber to the concentration of free Ca2+ in the cytosol that initiate contraction.

25.Name different component of membranous labyrinth.2 marks

Membranous Labyrinth:

It consists of three parts:

- Three semicircular canals
- Vestibule (sacculus and utricles)
- Cochlea
- The semicircular canals and vestibule are concerned with maintaining balance while cochlea is the sensory structure concerned with hearing.

26.What is reason behind latency period in action potential.

Latency Period in Coupling

It takes several milliseconds to begin contraction after the arrival of an AP.

This latency is because of the large size of skeletal muscle fibers which cannot contract unless action potential spreads deep into the fiber to the vicinity of each myofibril.

During this latent period, action potential is transmitted along the transverse tubules (T tubules) deep within the fiber.

27. What are the effect of neuromodulation.3

Effects of Neuromodulation:

The phenomenon of neuromodulation plays critical roles in the development of complex behavioral patterns e.g. happiness, revenge, reward, greed and exploration. Neuromodulation is also involved in the processes of thinking, cognition, planning, learning and memory.

Many behavioral problems are also based on neuromodulation e.g. mood swings, sleep disturbances, feelings of stress, anxiety, depression and anger.

28.Name different component involve in the mechanism of conductivity of heart.5 marks

The atrial internodal pathways that conduct impulse from the S-A node to the atrioventricular (A-V) node.

- The A-V node, in which the impulse from the atria is delayed before passing into the ventricles.
- The A-V bundle, which conducts the impulse from the atria into the ventricles.
- The left and right bundle branches of A-V bundle.
- Purkinje fibers, which branch off from bundle branches and conduct the cardiac impulse to all parts of the ventricles.

29.What type of cells are present in islet of lengerhens?

Islets of Langerhans

- The islets contain four types of cells:

1. Beta Cells

- constitute 60% of the Islets. They secrete insulin

2. Alpha Cells

- constitute 25% of the islets .They secrete glucagon

3. Delta Cells

- constitute 10% of islets. Secrete somatostatin that inhibits the secretion of insulin and glucagon.

4. PP Cells

- Present in small number. Secrete a hormone of uncertain function called pancreatic polypeptide.

30.Function of estrogen in pregnancy? 5

Functions of Estrogens in Pregnancy

During pregnancy, estrogens exert mainly a proliferative function on most reproductive and associated organs of the mother. They cause:

(1) enlargement of mother's uterus
(2) enlargement of mother's breasts and growth of breast ductal structure (3) enlargement of mother's external genitalia.

They also relax the pelvic ligaments of the mother, so that the pubic symphysis becomes elastic. These changes allow easier passage of the fetus through the birth canal.

31.Diff b/w cGMP and cAMP? 3

Differences b/w cGMP and cAMP Pathways

A few of such differences include:

- The production of cGMP is catalyzed by guanylate cyclase enzyme from GTP.
- Guanylate cyclase occurs in two forms: one bound to the plasma membrane and one free in the cytoplasm. In contrast, adenylate cyclase is always bound to the plasma membrane.
- Guanylate cyclase becomes active as the Ca_{2+} concentration is increased within the cell, while adenylate cyclase activity is increased when Ca_{2+} conc. is low.
- cGMP activates a specific protein kinase, protein kinase G instead of protein kinase A.

32. Pace marker and it's types.

Pacemaker

The pacemaker is the excitatory region which generates the rhythmical impulses that control the rhythmicity of cardiac chambers.

Pacemaker Types

Two basic types of pacemakers in animals with pumping hearts:

- **Neurogenic pacemakers**
- **Myogenic Pacemakers Neurogenic and Myogenic Pacemakers**

A. In many invertebrate hearts, the pacemaker is neurogenic that consists of neurons. These hearts are known as neurogenic hearts.
B. The pacemaker in some invertebrate and all vertebrate hearts is myogenic i.e. consists of specialized selfexcitatory muscle cells. Such hearts are known as myogenic hearts.

33. Gap junction. Writes its works

Gap Junctions

Electrical activity, initiated in the pacemaker region of the heart, spreads over the entire heart from one cell to another because the cells are electrically coupled via membrane gap junctions.

Gap junctions are regions of low resistance between cells and allow current flow from one cell to the next across intercalated disks.

Role of Gap Junctions

- Allow current flow from one cell to the next.
- Pacemaker's electrical activity spreads over the entire heart via gap junctions.

34. Note on axon.

Axon

- An axon is a long and thick process that arises from the cell body and has a constant diameter.
- The lengths of the axons vary from few millimeters to more than a meter.
- The longest axons are seen in whales, where the axon of a single spinal motor neuron may extend many meters from the base of the spine to the muscles in the tail fin.

Axon—Functions

- The axons are specialized to conduct signals away from the cell body.

They have evolved mechanisms that allow them to carry information for long distances with high fidelity and without loss.

Structural components of axons:

The axon contains a jelly-like semi-fluid substance called Axoplasm, surrounded by the plasma membrane which is called Axolemma.

Axon has mitochondria and ER but lacks Nissl's granules. So exons are not involved in protein synthesis.

Axon terminals:

- Each axon may divide into numerous branches at its termination. These branches of axon are called axon terminals.
- Each axon terminal has small extensions called telodendria with enlarged ends called terminal knobs. These knobs have granules or vesicles that contain neurotransmitters.
- The axon terminals allow transmission of signals to many other neurons, glands or muscle fibers simultaneously.
- **Axon hillock:**
- The region of an axon where it joins the cell body is cone-shaped and is called the axon hillock.
- This is the region where the signals that travel down the axon are generated.
- **Myelin sheath:**

• Specialized neuroglial cells, known as Schwann cells are located at regular intervals along the axons of many neurons.
• These Schwann cells secrete a fatty layer known as myelin sheath, over the axons.
• The myelin sheath is not continuous, but there are non-myelinated points in between, which are called Nodes of Ranvier.

35.Mammalian Ear decision.

The Mammalian Ear Ear

• Ear is the organ of hearing that detects sound within a particular frequency range.
• The human ear can detect sound frequencies lying between 20 to 20000 hertz.
• **Functional Anatomy of Mammalian Ear**

waves through three principally different media i.e., air, bone and fluid.
• The conducted waves are converted into electrical signals through a series of complex steps.
• Mammalian ear has 3 major divisions:
• External ear
• Middle ear
• Inner ear

The 3 parts are designed to conduct sound **External Ear:**
External ear includes:
• The pinna (auricle)
• External auditory meatus
• Tympanic membrane (eardrum) .

These structures act as a funnel to collect sound waves from environment
Sound is amplified and concentrated onto the eardrum that vibrates

Middle Ear

• Contains three auditory ossicles malleus, incus, stapes in a series.
• Malleus attached to eardrum and stapes to oval window of cochlea.

Functions of Middle Ear

1. Receive sound waves as vibrations from eardrum and transmit them onto oval window of cochlea.
    1. Amplify sound waves (more than 20 fold).

Inner Ear

• It consists of a bony labyrinth which is filled with perilymph and has an oval and a round window. • Oval window receives sound vibrations through stapes.
• In the bony labyrinth lies the membranous labyrinth that is filled with endolymph.

36.Myelin sheath function.

Myelin Sheath

• Myelin sheath is a thick and multi-layered membranous structure that surrounds many vertebrate axons.
• It is mainly composed of a fatty substance sphingomyelin.

Functions of Myelin Sheath: I. Electrical insulation
II. Increase the speed of conduction
III. Space efficiency
IV. Saltatory conduction **Electrical Insulation:**

• The sheath mainly contains lipid substance sphingomyelin that is a poor conductor of electrical current.
• It provides electrical insulation for the axon.
• It decreases ion flow about 5000-fold.
• So, it acts as an analogue to the plastic insulation around electrical wires.

Increase the Speed of Conduction:

• Insulation with myelin enhances the efficiency of longitudinal spread of current.
• So, it results in rapid transmission of nerve impulse.
• It renders thin axons higher conduction speed than very thick non-myelinated axons.
• e.g. 20 µm myelinated axon has faster speed than 40x thicker squid giant axon.
• **Space Efficiency:**
• Thousands of myelinated axons can be packed into space occupied by just one giant axon.
• **Saltatory Conduction:**
• Ions cannot flow through internodes that have thick myelin sheath, so APs are not generated in these regions.
• Ions flow only through the nodes of Ranvier.
• Inward Na+ current produced at a node during the rising phase of AP travels all the way to the next node.
• Here it depolarizes the membrane and regenerates the AP.
• AP appears to jump along the axon from node to node.
• This mechanism of conduction is called saltatory conduction.

37. sarcoplasmic reticulum (5marks).....

Sarcoplasmic Reticulum

- The sarcoplasmic reticulum (SR) is a network of membrane-bound tubules that extends throughout muscle cells on either side of a Z disk and extends from one Z disk to the next as well.
- In many features, it is similar to the endoplasmic reticulum in other cells.
- The SR has a special organization that is extremely important in controlling muscle contraction. • That is why the rapidly contracting types of muscle fibers have extensive network of sarcoplasmic reticulum.

38.sarcolemma and sarcoplasmic

The membrane of the cell is called sarcolemma, its cytoplasm is called sarcoplasm and its endoplasmic reticulum is known as sarcoplasmic reticulum.

39.Define cardiac and stroke volume ?2

Cardiac Output

Cardiac output is the quantity of blood pumped into the aorta each minute by the heart. This is also the quantity of blood that flows through the circulation.

The average cardiac output for the resting adult is often stated to be almost exactly 5 L/min.

Factors Affecting Cardiac Output

Cardiac output varies widely with the level of activity of the body.

The following factors directly affect cardiac output:

(1) Basic level of body metabolism

(2) Physical activity of the body

(3) Age

(4) size of the body

Stroke Volume

The volume of blood ejected by each beat of the heart is termed the stroke volume.

Stroke volume is the difference between the volume of the ventricle just before contraction (end-diastolic volume) and the volume of the ventricle at the end of a contraction (end-systolic volume).

40.Give the name hormone of adrenal cortex 3

A. Aldosterone
B. Cortisol,
C. corticosterone
D. cortisone

41.Exocrine glands (5 marks)

Exocrine glands

Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body.

Exocrine glands are more easily identified than endocrine glands because of their duct leading to the body surface. The fluid secretions may be either proteins (enzymes) or mucous or both.

Examples:

- Salivary glands produce saliva that is delivered to the oral cavity for partial digestion of food through parotid and submandibular ducts.
- Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.
- Lacrimal glands produce tears that is delivered through lacrimal duct on the surface of eye to provide lubrication.
- Mammary glands produce milk that is delivered through lactiferous ducts to the nipples for nourishing the young.

1. The main function of endocrine glands is to secrete hormones directly into the bloodstream 2. The Pineal Gland Runs a Daily Biological Clock.
3. The Thyroid and Parathyroid Glands Increase Metabolism and Regulate Calcium Levels.

42.hrmons secreted by ovaries

I. Estradiol(estrogen)
II. Progesterone

43.uses of kymograph 2 marks

Kymograph

A kymograph is a device that graphically records changes in the mechanical activities of animal tissues in the physiological experiments.

Uses of Kymograph

Since its invention in the 1840s, the kymograph has been used most commonly in the field of medicine to study various physiological and muscular processes.

It has been used to study the skeletal muscle contractions (twitch and tetanus) as well as the cardiac muscle activities (cardiac cycle).

It is also used to measure blood pressure and rate of respiration.

44.three main parts of ear 2 marks

Mammalian ear has 3 major divisions:

1. External ear
2. Middle ear
3. Inner ear

• The 3 parts are designed to conduct sound waves through three principally different media i.e., air, bone and fluid.

45.Sensory organs of humans.2 marks.

The major sensory organs of human body are eyes, ears, nose, tongue and skin.

46.Positive and negative feedback mechanism.3marks.

Positive feedback systems:

There are many biological processes that are regulated by positive feedback, although they are less commonly found. In positive feedback systems, an end product speeds up its production by enhancing the effect of original stimulus.

The figure explains the positive feedback system in a biochemical pathway. A product stimulates an enzyme in the reaction sequence, increasing the rate of production of the product.

Negative feedback systems:

In life, the most common form of regulation encountered is the negative feedback, in which accumulation of an end product works to stop or slow down that process.

47.Myosin molecules structure and functions 5 marks

Myosin Molecule • Each myosin molecule consists of two identical heavy chains which are coiled together to form a long tail. It also has two globular heads which are made from two heavy chains plus three or four calcium-binding light chains.

• The heads form cross bridges between the thick and the thin myofilaments during contraction.

Thin Filaments

• The thin filaments are 7-8 nm thick and extend across the I band.

• They are composed chiefly of actin molecules.

• Thin filaments also overlap myosin filaments in the peripheral darker regions of the A band. In these regions, six actin filaments surround each myosin filament while each actin filament is surrounded by three myosin filaments.

• In thin filaments, actin molecules are arranged in two chains which twist around each other.

• Two strands of another protein tropomyosin twist around the actin and help to stiffen it. In a relaxed muscle fiber, they block myosin binding so that the myosin heads can not bind to the thin filaments. • Thin filaments also have a three polypeptide complex troponin at intervals of about 40nm along the thin filament.

• One of the troponin polypeptides (TnI) is an inhibitory subunit that binds to actin, other (TnT) binds to tropomyosin and helps position it on actin while third (TnC) binds the calcium ions.

• Both troponin and tropomyosin help control the myosin-actin interactions involved in contractions.

The H Zone

The center of A band appears lighter than the other regions in a relaxed sarcomere.

- This region is called H zone and contains only thick filaments. There are no overlaps between the actin and myosin in this region. • The H zone is bisected by a dark line, the M line which contains enzymes important in energy metabolism.

48.ca ion role in excitation .

Hair cell excitation due to inward K_+ current also causes the opening of voltage-gated calcium channels. The resultant Ca_{2+} influx causes transmitter (mainly glutamate) release from the basal end of the cell onto the auditory nerve endings, stimulating them to send an electrical signal along the cochlear nerve.

49.Three growth hormons.

1. Piturity
2. Placenta
3. hypothalamus

50.Two vascular resistance

Vascular resistance is the resistance that must be overcome to push blood through the circulatory system and create flow.

- systemic vascular resistance (SVR
- total peripheral resistance (TPR)

51.What is myosin how it helps in muscular contraction

• When a muscle contracts, the thin actin filaments slide b/w the thick myosin filaments & move closer to the center of sarcomere. As a result, sarcomere becomes shorter.

52.Group of hormones insoluble in lipids.3 marks

The second messengers involved in signal transduction fall into three distinct groups:

- Cyclic nucleotide monophosphates e.g. cAMP (cyclic adenosine monophosphate) and cGMP (cyclic guanosine monophosphate).
- Inositol phospholipids e.g. inositol trisphosphate (InsP3) and diacylglycerol (DAG)
- Ca_{2+} ions and associated calmodulin

53.hair cilia

cilia (plural), are small hair-like protuberances on the outside of eukaryotic cells. They are primarily responsible for locomotion, either of the cell itself or of fluids on the cell surface.

54.Difference Pacinian corpuscles and meissners corpuscles.

Meissner's Corpuscles: are the touch receptors. They have encapsulated nerve endings which lie in papillae which extend into the ridges of fingertips.
Sense of touch perceived through skin having Meissner's corpuscles
Pacinian corpuscles: are situated quite deep in the skin. They have encapsulated nerve endings and receive deep pressure stimulus.
Pacinian corpuscles that are situated deep in the body and receive pressure stimulus. In the limbs, they receive vibrations.

55.myofilament and its types 3markss

Each myofibril is composed of myofilaments. Myofilaments are of two types, thin filaments and thick filaments. Thin filaments are composed of actin while the thick filaments are composed of myosin molecules.

56.What is meant by sensation?

Sensations arise when sensory receptor cells, receiving a stimulus, transmit signals through the nervous system to particular parts of the brain that interprets or perceives these signals. The subjective description of these neuronal perceptions is termed as sensation.

Exocrine Glands

Exocrine glands Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body. Exocrine glands are more easily identified than endocrine glands because of their duct leading to the body surface. The fluid secretions may be either proteins (enzymes) or mucous or both.

Examples:

- Salivary glands produce saliva that is delivered to the oral cavity for partial digestion of food through parotid and submandibular ducts.
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a) The main function of endocrine glands is to secrete hormones directly into the bloodstream
b) The Pineal Gland Runs a Daily Biological Clock.
c) The Thyroid and Parathyroid Glands Increase Metabolism and Regulate Calcium Levels.

Labeled Line Coding

Each receptor subtype for five kinds of taste sensations is connected to a particular set of axons. In that arrangement, for example, information about "sweetness" would be carried by somespecific subset of axons. Such a pattern is called labeled line coding.

Gastro Intestinal Hormones

Gastrointestinal hormones glucagon, gastrin and cholecystokinin, SubstanceP, endorphins, enkephalins and many other amino acid derivatives.

the ciliary muscles relax, lens becomes more rounded. This focuses the objects close to the eye.

Mammalian Ear Decision.

The Mammalian Ear

- Ear is the organ of hearing that detects sound within a particular frequency range.
- The human ear can detect sound frequencies lying between 20 to 20000 hertz**.**

Functional Anatomy of Mammalian

Ear waves through three principally different media i.e., air, bone and fluid.

 The conducted waves are converted into electrical signals through a series of complex steps.

Mammalian Ear Has 3 Major Divisions:

- External ear
- Middle ear
- Inner ear

The 3 parts are designed to conduct sound External Ear:

External ear includes:

- The pinna (auricle)
- External auditory meatus
- Tympanic membrane (eardrum) .

These structures act as a funnel to collect sound waves from environment Sound is amplified and concentrated onto the eardrum that vibrates

Middle Ear

- Contains three auditory ossicles malleus, incus, stapes in a series.
- Malleus attached to eardrum and stapes to oval window of cochlea.

Functions of Middle Ear Receive sound waves as vibrations from eardrum and transmit them onto oval window of cochlea.

 Amplify sound waves (more than 20 fold).

Inner Ear

- It consists of a bony labyrinth which is filled with perilymph and has an oval and a round window.
- Oval window receives sound vibrations through stapes.
- In the bony labyrinth lies the membranous labyrinth that is filled with endolymph.

Hrmons Secreted By Ovaries

I. Estradiol(estrogen)
II. Progesterone

Three Main Parts Of Ear

Mammalian ear has 3 major divisions:

1. External ear
2. Middle ear
3. Inner ear

The 3 parts are designed to conduct sound waves through three principally different media i.e., air, bone and fluid.

Components Of Rhodopsin

A rhodopsin molecule consists of two major components: Opsin protein and a light-absorbing molecule that may be retinal or 3- dehydroretinal. Retinal is the aldehyde of a carotenoid vitamin A1 (Retinol). 3- dehydroretinal is the aldehyde of vitamin A2 (3-dehydroretinol). In addition to these major

components, rhodopsin includes a sixsugar polysaccharide chain and a variable number (as many as 30 or more) of phospholipid molecules.

Color Blindness (Red & Green)

Color blindness is caused due to a mutation in one of the cone opsin genes, resulting in absence of one type of pigmented cones. A person missing a single type of color receptive cones is unable to distinguish some colors.

Red-Green Color Blindness

The green, yellow, orange, and red colors between the wavelengths of 525 and 675 nm, are normally distinguished from one another by the red and green cones. If either of these two cones is missing, the person cannot distinguish these four colors. The person is especially unable to distinguish red from green and is therefore said to have red-green color blindness.

Thermoreceptors Are Found In Skin

Thermoreceptors that keep track of thermal state of the body. They are found scattered on the skin.

Hair Cell Cilia Types

- Hair cells are named for the many cilia that project from the apical end of each cell.
- These cilia fall into two classes: kinocilium and stereocilia.

Thermoreceptor Types

Warmth and Cold Receptors There are two kinds of thermoreceptors in the external skin and upper surface of the tongue:

1) Cold receptors
2) Warmth receptors

Both types of receptors are quite sensitive and enable humans to detect a change in skin temp. of as little as 0.01°C.

Cold Receptors

The cold thermoreceptors are 3.5 times more common in skin than heat receptors. They consist of free nerve endings of neurons that have thin myelinated Aδ fibers having faster conduction velocity (19m/s). They increase their firing rate when the skin is cooled below body temperature.

Warmth Receptors

Warmth Receptors increase their firing rate in response to temperatures above body temperature. They consist of free nerve endings of neurons that have unmyelinated C fibers with low conduction speed (0.8m/s). Both these receptors are quite sensitive which enable human beings to detect a change in skin temperature of as little as 0.01°C.

Hormones Released By Thyroid Glandular

Cells Secretion of T3 and T4

The secretion of T4 predominates T3. However the target cells convert most of T4 to T3, which is the active form and carries out the major functional roles. The secretion of these hormones occurs by the stimulation of hypothalamus that releases TSH-releasing hormone (TRH). The stimuli for the release of TRH are stress, cold, low skin temperature and low metabolic rate. TRH acts on anterior pituitary gland that releases thyroid stimulating hormone (TSH). TSH causes the release of thyroid hormones.

Functions Of Catecholamines

Catecholamines help the body respond to stress or fright and prepare the body for "fight-or-flight" reactions. The adrenal glands make large amounts of catecholamines as a reaction to stress. The main catecholamines are epinephrine (adrenaline), norepinephrine (noradrenaline), and dopamine.

Hypothyroidism

Hypothyroidism usually results from the lack of dietary iodine. It is characterized by two types of diseases:

Cretinism

Iodine deficiency during early stages of development results in cretinism.In cretinism somatic, neural and sexual development is severely retarded, metabolic rate and resistance to infection is reduced.

Goiter

Inadequate production of thyroid hormones in adults leads to excessive production of TSH.

Function Of Luteinizing Hormone In Male & Female?

LH stimulates production of estrogen and progesterone by the ovary. It causes ovulation and formation of the corpus luteum.

In males, LH is also called as interstitial cell–stimulating hormone (ICSH) and stimulates testosterone production by the Leydig cells of testes

Lobes Of The Pituitary Gland

The pituitary gland is composed of three lobes:

- Anterior lobe (adenohypophysis)
- Intermediate lobe (pars intermedia)
- Posterior lobe (neurohypophysis)

Anterior Lobe (Adenohypophysis) The anterior pituitary is fleshy, glandular and highly vascularized in all animals. It contains five types of glandular cells which are controlled by regulatory hormones of the hypothalamus. On stimulation of hypothalamic releasing hormones, they synthesize and secrete six hormones. Posterior lobe (neurohypophysis) The posterior pituitary is non-fleshy and nonglandular. It has neural composition and is considered as an extension of the hypothalamus.

What Is Myosin How It Helps In Muscular Contraction

When a muscle contracts, the thin actin filaments slide b/w the thick myosin filaments & move closer to the center of sarcomere. As a result, sarcomere becomes shorter.

Group Of Hormones Insoluble In Lipids.

The second messengers involved in signal transduction fall into three distinct groups:

- ✓ Cyclic nucleotide monophosphates e.g. cAMP (cyclic adenosine monophosphate) and cGMP (cyclic guanosine monophosphate).
- ✓ Inositol phospholipids e.g. inositol trisphosphate (InsP3) and diacylglycerol (DAG)
- ✓ Ca2+ ions and associated calmodulin

Myofilament And Its Types

Each myofibril is composed of myofilaments. Myofilaments are of two types, thin filaments and thick filaments. Thin filaments are composed of actin while the thick filaments are composed of myosin molecules.

Hormones In Regulation Of Water And Ions

These hormones include:

- Antidiuretic hormone (ADH)
- Aldosterone
- Atrial natriuretic Hormone
- Calcitonin
- Parathormone

The Signaling System Of Camp

Many hormones use the adenylate cyclase–cAMP second messenger system to stimulate their target tissues. The hormone binds with the receptor that is coupled to a G protein. The G protein stimulates the membrane-bound adenylate cyclase enzyme. Activated adenylate cyclase catalyzes the conversion of adenosine triphosphate (ATP) into cAMP in the cytoplasm. The cAMP then activates an enzyme called cAMP-dependent protein kinase. This enzyme phosphorylates specific proteins in the cell, triggering a cascade of biochemical reactions that ultimately lead to the cell's response to the hormone.

Muscle Twitch And Tetanus .

The main difference between a skeletal muscle twitch and tetanic contraction is that twitch occurs only momentarily while tetanic contraction occurs in a prolonged and steady state. ... The contractions in muscle twitch are also light while tetanic contraction is characterized by strong contract.

Gap Junction.

Writes its works Gap Junctions Electrical activity, initiated in the pacemaker region of the heart, spreads over the entire heart from one cell to another because the cells are electrically coupled via membrane gap junctions. Gap junctions are regions of low resistance between cells and allow current flow from one cell to the next across intercalated disks.

Role of Gap Junctions

- Allow current flow from one cell to the next.
- Pacemaker's electrical activity spreads over the entire heart via gap junctions.

Tetanus And Types

A tetanic contraction is a sustained muscle contraction evoked by stimulation from simultaneous multiple impulses. Each stimulus causes a twitch.

Types of Tetanus

A tetanic contraction can be either unfused (incomplete) or fused (complete).

Unfused Tetanus

An unfused tetanus is when the muscle fibers do not completely relax before the next stimulus because they are being stimulated at a faster rate.

Fused Tetanus

Fused tetanus is when there is no relaxation of the muscle fibers between stimuli and the twitches overlap. It occurs during a high frequency of stimulation. A fused tetanic contraction is the maximal possible contraction. During tetanized state, the contracting tension in the muscle remains constant in a steady state

Sensory Receptors Of Cardiovascular System

The body employs a variety of sensory receptors for monitoring the status of cardiovascular system.

1) **Baroreceptors:** monitor blood pressure at various sites in the cardiovascular system
2) **Chemoreceptors:** monitor CO2, O2, and pH of the blood.
3) Cardiac mechanoreceptors
4) Thermoreceptors

Laminar Flow Or Streamline Flow In Pipes (Or Tubes)

Occurs when a fluid flows in parallel layers, with no disruption between the layers. ... Turbulent flow is a flow regime characterized by chaotic property changes. This includes rapid variation of pressure and flow velocity in space and time.

FINAL TERM

1. What is blubber and its role 2 marks

Blubber is a thick layer of fat, also called adipose tissue, directly under the skin of all marine mammals.
Role: As fatty tissues are little metabolically active & have little blood vascularization so it is a good insulator b/c of low thermal conductivity and prevent heat loss through body surface.

Marine mammals (seals, walruses, sirenians, cetaceans—whales) have thick layers of insulating subcutaneous fat which is known as blubber.
Blubber is a good insulator because it has a lower thermal conductivity.

Fatty tissues are metabolically very inactive and have little blood vascularization. This prevents heat loss through body surface.

2. Define Peristaltic 2 marks

Waves of contraction and relaxation in the smooth muscles present in the walls of alimentary canal. The alimentary canal starts from oral cavity and end at distal end of rectum. Esophagus conducts the swallowed bolus by peristaltic movement.
The esophagus conducts the bolus by peristaltic movement.
Peristaltic movements are the waves of contraction and relaxation in the smooth muscles of alimentary canal walls.
These movements s
tart from oral cavity and end at the distal end of rectum.

3. Write down the basic types of alimentary system 2.marks

Types of Alimentary Systems:
Two basic plans of alimentary systems in animals:

A. Sac-like gastro vascular cavity
B. Tubular alimentary canal or gut

4. How alveolar collapse is prevented? 2 marks

Prevention of Alveolar Collapse: Alveolar collapse normally does not occur for two reasons
1. Surrounding tissue prevents over-expansion of alveoli.
2. Presence of pulmonary surfactants.

5. Differentiate between laminar and turbulent flow? 2marks

Laminar Flow:
Flow is streamlined and continuous. The velocity of fluid is constant at every point. Flow occurs in layers at different velocities. Flow is zero at the wall and maximal at the center, A pressure difference supplies the force required to slide adjacent layers to past each other.
Turbulent Flow:
Direction of fluid movement is not aligned. It requires higher energy to move blood through Vessels. Turbulence also happens if smoothness is reduced by any obstruction e.g. Buildup of fatty deposits on arterial walls.

6. Name mechanoreceptor fiber in the arterial wall 2 marks

Atrial Receptors: Atrial walls have mechano-receptive afferent fibers including:

A. Myelinated A-type afferent fibers
B. Myelinated B-type afferent fibers
C. Unmyelinated C-type afferent fibers
D. Stretch-sensitive secretory cells

7. Fahraeus- lindqvist effect 2 marks

Fahraeus-Lindqvist Effect: Phenomenon of plasma skimming & resultant increased blood flow velocity is called Fahraeus Lindqvist effect. It reduces energy required to drive blood through microcirculation

8. Types of vascular resistance 2 marks

Vascular Resistance "The resistance that must be overcome to push blood through the circulatory system and create flow"
Types

1) Systemic vascular resistance (SVR)
2) Pulmonary vascular resistance (PVR)

9. Objectives of ethology 3 marks

Ethology:
"Scientific study of animal behavior in natural environment".
Objectives of Ethology: Involves understanding of

1. Understanding the stimuli that elicit behavior.

2. Understanding the physiological mechanisms that mediate the response.

3. Understanding how an animal's experience during growth and development influence the response.

4. Understanding how the behavior aids in survival and reproduction.

5. Understanding the behavior's evolutionary history.

10. Four accessory glands of elementary canal? 3 marks

Accessory Glands: Secrete digestive juices through ducts.
In mammals, the alimentary canal is associated with various accessory glands that secrete digestive juices through ducts into the gut.
The accessory glands of the mammalian digestive system are:

a. Three pairs of salivary glands
b. Gastric glands
c. Pancreas
d. Liver
e. Gallbladder

11. Briefly describe ornithine urea cycle 3 marks

Ornithine-Urea Cycle
• All vertebrates except most teleost fishes, synthesize urea primarily in the liver via the ornithine-urea cycle.
• In this cycle, two ammonia and one CO2 molecule are added to ornithine to form arginine.
• Arginine is cleaved by the enzyme arginase, to remove a molecule of urea and regenerate ornithine.

12. Factors affects systemic vascular resistance 3 marks

Factors affecting SVR
Elasticity of vessel wall
Diameter of vessel
Decreasing vessel diameter (vasoconstriction) increases SVR
Increasing vessel diameter (vasodilation) decreases SVR
Vascular resisance is determined mainly by the muscle tone in the smooth muscle tissue of the tunica media and the elasticity of the elastic fibers there.
Vasoconstriction (i.e., decrease in blood vessel diameter) increases SVR, whereas vasodilation (increase in diameter) decreases SVR.

13. 3 uses of crop in various animals. (3) marks

Crop
• A sac-like expanded section in esophagus of some animals. Used to store food without digesting.
Crop plays various roles in different animals. For example
• Food storage without digestion

• Fermenting the food
• Prepare food for nestlings (pigeon’s milk)
In some animals crops are also used to ferment or digest foods for purposes other than normal digestive process. For example some parent birds prepare food in this way to be regurgitated for their nestlings e.g. pigeon’s milk.

14. Name 3 vasodilators in the body? 3 marks

1. Histamine released from connective tissue and white blood cells in injured tissues causes vasodilation
2. Plasma kinins (e.g. bradykinin) are potent vasodilators in damaged tissues
3. Nitric oxide (NO) serves as a major inducer of vasodilation in cardiovascular system also causes relaxation of vascular smooth muscle and acts as vasodilator
4. Lactic acid generated by tissue cells
5. Rising conc. of K+ or H+ ions in interstitial fluid
6. Prostacyclin causes vasodilation and acts as an anticoagulant.

15. Write the name of local vasoconstrictor? 3 marks

1. Prosta gland and thromboxanes released by activated platelets and white blood cells.
2. Endotheline released by damaged endothelial cells.
   If autoregulation fails to normalize conditions at the tissue level, neural mechanisms and endocrine factors are activated.

16. Write down the composition of the Bile? 3 marks

Bile: Vertebrate liver secretes bile. Bile doesn't contain any digestive enzymes It contains many salts Bile salts are essential for the digestion of fats
Composition of Bile:
Bile consists of water and a weakly basic mixture of cholesterol, lecithin, inorganic salts, organic salts, and bile pigments.

The bile organic salts are manufactured by the liver from cholesterol and amino acids complexed with sodium.

The bile pigments are derived from biliverdin and bilirubin, which are products of the breakdown of hemoglobin coming from ruptured red blood cells.

17. What is Rete Mirabile? Mention its function in the Tuna fish? 5 marks

Rete Mirabile A highly evolved system of countercurrent heat exchange. Found widely in homeothermic and heterothermic vertebrates. The artery carrying warm blood toward extremity is completely encased in a circlet of veins carrying back cold blood.

Function in the Tuna fish:

Forms an extensive heat exchange surface b/w inflowing and outflowing blood. Large number of rete mirabile in tuna (fish) are used to regulate temperature of brain, muscles, and eyes. Swimbladders of some fishes (e.g. eels) are associated with rete mirabile and functioning as CO2 exchangers.

A countercurrent network arrangement of arterial and venous capillaries is referred to as a rete mirabile. Before entering a tissue, an artery divides into a large number of small capillaries that parallel a series of venous capillaries leaving the tissue.

The arterial capillaries are surrounded by venous capillaries.

Rete Mirabile—Functions

This arrangement forms an extensive exchange surface between inflowing and outflowing blood.

Examples

Tuna (fish) have a large number of rete mirabile, which are used to regulate the temperature of the brain, muscles, and eyes.

The rete mirabile leading to the swimbladder of some fish (e.g. eels) function as a carbon dioxide countercurrent exchanger.

18. Note on physiological adaptation of ectothermic in hot environment 5 marks

Physiological Adaptations

• Ectotherms have high heat conductance
• Radiate heat rapidly—by moving to less warmer places
• Panting: Above 40oC many reptiles start panting—like birds and mammals
• Increases heat loss through respiratory evaporation, causing cooling.
• Regulating heart rate and flow of blood to surface tissues—to eliminate heat (e.g. marine iguanas)
• Gila monster (Heloderma suspectum) evaporates water from its cloaca to cool—a process similar to sweating in mammals

Most ectotherms have high heat conductance that allows them to radiate heat rapidly on moving to less warm places.

Above 40oC many reptiles start panting, just like birds and mammals to increase heat loss through respiratory evaporation that causes cooling.

The marine iguanas can regulate heart rate and flow of blood to the surface tissues to either absorb or eliminate heat.

The Gila monster (Heloderma suspectum) evaporates water from its cloaca to cool its internal body temperature—a process similar to sweating in mammals.

19. Write a note on types of stomach in vertebrates 5 marks

Types of Stomach in Vertebrates: Based on number of chambers, stomachs are classified as:

a. Monogastric stomachs
b. Digastric stomachs

Monogastric Stomachs

A monogastric stomach consists of a single strong muscular tube or sac.

Carnivorous and omnivorous vertebrates have a monogastric stomach.

Digastric Stomachs

Digastric stomachs are multichambered and are found in ruminant mammals of order Artiodactyla e.g. deer, giraffe, sheep, cattle, camel, llama etc.

These stomachs have four chambers which are separated into two divisions.

• The first division consists of fermentation chambers called rumen and reticulum.
• The second division comprises the omasum and abomasum (true stomach).

20. What is hypoxia? Responses of aquatic animals in it 5 marks

Hypoxia • "A condition in which the body or a region of the body is deprived of adequate oxygen supply" **Hypoxia in Aquatic Environment**
• O2 levels in aquatic environment vary more frequently
• Aquatic animals are faced with frequent and rapid changes in oxygen levels. Many aquatic animals can withstand very long periods of hypoxia
Respiratory Responses of Aquatic Animals to Hypoxia

Oxygen level in aquatic environment varies frequently. Aquatic animals are subjected to more frequent and rapid changes in oxygen levels.

Many aquatic animals can withstand very long periods of hypoxia. Many of these animals utilize a variety of anaerobic metabolic pathways to survive the period of reduced oxygen availability.

Many animals adjust the respiratory and cardiovascular systems to maintain oxygen delivery during reduced oxygen availability. For example, many fishes increase gill ventilation rate in hypoxic conditions. Ram ventilators (e.g. tuna) swim forward with their mouth open, increase the size of gap during hypoxia to increase water flow over the gills.

21. What are pheromones? Also describe their Role? 10 marks

Pheromones:
• Chemical messengers released by many animals in environment to communicate. Spread through air.
Characteristics:

a. Affect the behavior of individuals of a species
b. Effective at remarkably low concentrations
c. Have particular odors
d. Detected by olfactory receptors of receiving animal

Roles of Pheromones:

1. Recognize Members of Species:
Unicellular organisms recognize members of their own species. Many insects identify members of their colony.
2. Reproductive Roles:
Pheromones attract members of opposite sex for breeding.
Induce courtship behaviors in members of a species.
Induce spawning behavior in aquatic animals with external fertilization.

3. Territorial Behaviors:
Many male mammals mark their territories with pheromones.
It warns other males of their occupied area.
4. Insect's Social Order:
In a honeybee colony, pheromones produced by queen maintain hive's complex social order.
5. Alarm Calls:
Members of some species produce pheromones on getting injured.
It warns other members by the presence of danger in area.
6. Repel Predators:
Pheromones produce by the pray may repel predators.
Example: The skunks produce foul-smelling musk that makes skunks unpalatable or distasteful to their enemies.

22.Define vestibular apparatus ? 2marks

Vertebrate Organ of Equilibrium
• Called as vestibular apparatus, located in the inner ear. It has:

1. • Saccule
2. • Utricle
3. • Semicircular canals

23.Abnormalities due to GH? 5 marks

Disturbances in the secretion of GH lead to abnormal growth and development in humans:

Gigantism
• Hypersecretion of GH before puberty causes excessive size and stature.
Developmental
Abnormalities Due to GH
Acromegaly
• Hyper secretion of GH after maturity causes enlargement of bones of the head and limbs.
Dwarfism

- Insufficient secretion of GH during childhood and adolescence causes underdevelopment of The body (short stature).

24.Oxytocin? 3 marks

Is a type of harmone .

oxytocin are released by axon terminals in the posterior pituitary. Stimulates milk ejection, uterine contractions during child birth.

Neurohypophyseal hormones are synthesized and packaged in the cell bodies of two groups of neurosecretory cells in the anterior portion of the hypothalamus:

1. Supraoptic nuclei
2. Paraventricular nuclei

25. Five types of hormone ?

Based on their structure and pathway for synthesis, hormones are divided into four groups:

- Peptide and protein hormones
- Amine hormones
- Steroid hormones
- Prostaglandins

26.Gastrointestinal track? 5

Gastrointestinal tract

Cholecystokinin Stimulates secretion of enzymes by pancreatic acinar cells
Stimulates gall bladder contraction
Chymodenin Stimulates secretion of chymotrypsinogen from pancreas
Gastrin Stimulates HCl secretion in stomach
Secretin Stimulates secretion of bicarbonate
Substance P Acts as enteric neurotransmitter for feeling of pain
Motilin Increases motility of intestinal villi

27. Prostaglandins.10 marks

Prostaglandins are about 16 hormone-like local regulators substances that constitute a family of

cyclic, long-chain, unsaturated, hydroxy fatty acids.

Synthesis

Prostaglandins are produced by all or nearly all tissues. They are synthesized in membranes from arachidonic acid, which is produced by cleavage of membrane phospholipids by phospholipases.

Mode of Action

In some cases, they act locally as paracrine agents and in other cases they act on distant target

tissues in endocrine fashion. They have a rapid, short-lasting effect, similar to that of lipid-insoluble hormones.

They bind to cell-surface receptors linked to the cAMP pathway.

Functions

Prostaglandins have diverse actions on a variety of tissues, particularly involving smooth muscles.

1. Role in Fertilization

Prostaglandins present in semen aid in fertilization by reacting with the female cervical mucus to make it more receptive to sperm movement. They also stimulate the smooth muscles of the female's uterine wall to contract, helping sperm reach an egg.

2. Aid During Labor

Certain prostaglandins are secreted by the placenta at the onset of childbirth. They make the muscles of uterus more excitable, enhancing uterine contractions during labor.

3. Role in Immune Response

The damaged tissues, that induce immune response, produce prostaglandins. Some prostaglandins act as local regulators of inflammation, promoting fever and inflammation and also intensify the sensation of pain. They worsen pain by increasing nociceptor sensitivity to noxious stimuli. The interleukin-1, produced in immune response, causes fever by inducing the formation of a prostaglandin E2 which acts in the hypothalamus to

elicit the fever reaction. The anti-pyretic, anti-inflammatory and pain-relieving effects of aspirin and ibuprofen are actually due to the inhibition of prostaglandin synthesis by these drugs.

4. Blood Clot Formation

Prostaglandins also help the platelets to aggregate and form blood clots. This is the basis of use of aspirin by patients at risk for a heart attack due to formation of clots.

5. Stomach Lining

Prostaglandins also help maintain a protective lining in the stomach.As aspirin interferes with prostaglandin synthesis, long-term aspirin therapy can result in damage to this protective lining, causing debilitating stomach irritation.

6. Production of Erythropoietin

Several prostaglandins stimulate the production of erythropoietin by kidneys. Erythropoietin is a hormone that stimulates the production of erythrocytes.

28.neuromodulation?

A physiological process by which neurotransmitters released by a small group of neurons diffuse through large areas of the CNS and interact and modulate the effect of many neurons simultaneously.

- This is in contrast to the normal synaptic transmission. Neurotransmitters involved in altering the cellular properties and efficacy of synaptic transmission of multiple postsynaptic neurons.
- They act through Gprotein linked receptors that use voltage-gated ion channels for ion movements.
- Generation of response is slow, but lasts longer.

29.role of corpus leutum in pregnancy

If the egg is fertilized and implants, active corpus luteum is maintained.

- So progesterone and estrogen secretion continues until placenta fully develops to take over the production of these hormones.
- At this stage, corpus luteum degenerates. In many mammals, (e.g. rat) corpus luteum is stimulated by prolactin and continues to grow and secrete estrogen and Progesterone throughout the gestation period.

30.Types of stimuli

Two types:

- External stimuli Stimuli of odor, touch, light, sound and gravitation
- Internal stimuli Stimuli of pain, homeostatic imbalances and blood pressure

31.taste buds and taste receptors

The gustatory organs ofvertebrates are called taste buds.

A taste bud is composed of about 50 types of modified epithelial cells, including:

- Supporting cells (sustentacular cells)
- Basal cells
- Taste receptor cells
- basal cells are progenitor cells that give rise to new
- They regularly generate new sensory taste receptor cells which have an active life of only 10 days.

32.Feedback Mechanism (2)

Mechanism of self-regulation of Biological processes. Output or product itself regulates the process Maintenance of homeostasis

33.Acclimatization with example (2)

Acclimatization is a physiological, biochemical, or anatomic change within
an individual animal that results from the animal's chronic exposure to new, naturally occurring environmental conditions.
Example.
Let's consider an animal that voluntarily migrates from a valley to a high mountain i.e. a voluntary change happens in its natural environment in which oxygen partial pressure is

low.

34.Adrenal cortex hormone (3)

Adrenal (Cortex)

Aldosterone Increases blood Na+levels; increase K+secretion

Cortisol, corticosterone, cortisone

Role in carbohydrate metabolism increase blood glucose levels anti-inflammatory effects

35.Parts of eye take part in focusing (3)

The parts of the eye involved in focusing and image formation are:

• Cornea, a biconvex lens, pupil and retina.

36.Role of T3 and T4.10 marks

Metabolic Roles of T3 & T4

• They regulate bioenergetics by stimulating cellular respiration, oxygen consumption and metabolic rate.

• The acceleration of metabolism leads to a rise in heat production. This is of major importance in the thermoregulation of many vertebrates.

• They also sensitize some tissues to epinephrine that helps to maintain normal blood pressure, heart rate and muscle tone.

• They also promote normal motility of gastrointestinal tract and regulate reproductive

functions.

37.Acclimatization2 marks

An adaptive change within an individual animal.

• Happens due to chronic exposure to new, naturally occurring environmental conditions.

38.Rhodopsin 2 marks

The visual pigment in the rods is rhodopsin. Rhodopsins absorb photons of light energy and produce a generator potential.

Opsins are coupled to light-absorbing photopigment Retinal.

• This form a functional pigment molecule rhodopsin.

39.define line coding 2 marks

Each receptor subtype for taste sensations is connected to a particular set of axons in the nerve.

• In such an arrangement, information about one taste e.g. "sweetness" would be carried by some specific subset of axons.
• Such a pattern is called labeled line coding.

40. Explain the types of hair cells

Hair cells are ciliary cells.

• Found in several sensory organs of vertebrates.
• They are highly sensitive mechanoreceptors.
• Responsible for transducing mechanical stimuli into electrical signals.

Systems Based on Hair Cells: Examples

• Lateral-line system of fishes and amphibians, involved in detection of motion in the surrounding water.
• This system is based on hair cells.

Hair Cell Cilia

• Many cilia project from the apical end of each cell (reason of naming).
• Cilia are of two types:
• Kinocilium • Stereocilia
• A hair cell has a single kinocilium.
• Kinocilium has a "9 + 2" arrangement of internal microtubules (similar to other motile cilia).
• Each hair cell has 20-300nonmotile stereocilia.
• Stereocilia are structurally and developmentally distinct from the kinocilium.
• They are not formed of microtubules.
• They are formed of actin filaments.

• The stereocilia are arranged in order of increasing length from one side of the cell to the other.

41.diseases by hypothyroidism

Characterized by low production of throxine.

• It results from the lack of dietary iodine.
• It is represented by two types of diseases:
• Cretinism
• Goiter
• Cretinism results from iodine deficiency during early stages of development. In cretinism somatic, neural and sexual development is severely retarded, metabolic rate and resistance to infection is also reduced.
• Inadequate production of thyroxine in adults leads to goiter. In goiter, TSH is produced excessively that causes overstimulation of thyroid gland resulting in its enlargement
(hypertrophy).

42.Name glandular cells of thyriod and name their hormones too. (3marks)

Glandular Cells and Hormones of Adenohypophysis

1. Somatotropes—produce growth hormone (GH)
2. Corticotropes—produce adrenocorticotropic hormone (ACTH)
3. Thyrotropes—produce thyroid-stimulating hormone (TSH)
4. Gonadotropes—produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH)
5. Lactotropes—produce prolactin (PRL)

43.Range fractionation

In a sense organ, sensory receptors are arranged in an order of increasing sensitivity to different range of intensities of the stimulus.

• This hierarchical arrangement of receptors is known as range fractionation. Each individual receptor, in a sense organ, covers only a fraction of the total dynamic range of the stimulus intensities.
• Receptors work together in a hierarchical way to provide discrimination of stimulus intensities.

44.Placental hormone

In pregnancy, placenta secretes large quantities of hormones, essential for maintenance of pregnancy.

• These hormones include:
• Human chorionic gonadotropin
• Estrogens
• Progesterone
• Human chorionic somatomammotropin

45.Progesteron

Progesterone is secreted in tremendous quantities by placenta.

• This secretion is about 10-fold more than normal levels. Causes development of decidual cells in endometrium. These cells play an important role in the nutrition of the early embryo.
• It decreases contractility of the uterus to prevent spontaneous abortion.

46.Color blindness 3

Color blindness is caused due to a mutation in one of the cone opsin genes that results in the absence of one type of pigmented cones.

• A person missing a single type of color receptive cones is unable to distinguish some colors.

47.Three use of crop?

A sac-like expanded section in esophagus of some animals. Used to store food without digesting. • Generally found in infrequent or quick feeders • Allows quantities of food to be stored for digestion at a later time

48.Define peristaltic?

The wavelike muscular contractions in tubular structures, especially organs of the digestive system such as the esophagus and the intestines. **Peristalsis** is characterized by alternate contraction and relaxation, which pushes ingested food through the digestive tract towards its release at the anus.

49.Objective of ethology?

Ethology is the scientific and **objective** study of animal behaviour, usually with a focus on behaviour under natural conditions, and viewing behaviour as an evolutionarily adaptive trait.

50.Three local vasodilator?

Vasodilation is the widening of blood vessels. It results from relaxation of smooth muscle cells within the vessel walls, in particular in the large veins, large arteries, and smaller arterioles. ...

 sildenafil (Viagra)
 tadalafil (Cialis)

 vardenafil (Levitra)

51.Three respiratory pigments?

A **respiratory pigment** is a molecule, such as hemoglobin in humans and other vertebrates, that increases the oxygen-carrying capacity of the blood. The four most common invertebrate **respiratory pigments** are hemoglobin, hemocyanin, hemerythrin and chlorocruorin.

52.Differentiate between Norepinephrine and Epinephrine?2

Norepinephrine is the primary excitatory transmitter in postganglionic cells of sympathetic system.

Epinephrine is excitatory at some synapses and inhibitory at others depending on postsynaptic membrane

53.Define vestibular apparatus?

The vertebrate organ of equilibrium is the vestibular apparatus. Vestibular apparatus, located in the inner ear. It has:

..The saccule

• The utricle

• Three semicircular canals lying orthogonally in three mutually perpendicular planes.

54.Name two components of rhodopsin ? 2mks

A rhodopsin molecule consists of two major components:

Opsin protein

A light-absorbing molecule (retinal or 3dehydroretinal) Function of LH in male and female.

In Females LH stimulates production of estrogen and progesterone by ovary.

It causes ovulation and formation of corpus luteum.

In males, LH is also called as ICSH (interstitial cell stimulating hormone) It stimulates testosterone production by the Leydig cells of testes.

55.What are gonadotropin

Gonadotropins: Luteinizing and Follicle Stimulating Hormones. Luteinizinghormone (LH) and follicle-stimulating hormone (FSH) are called gonadotropinsbecause stimulate the gonads - in males, the testes, and in females, the ovaries.

56.Define the terms gland and secretion? 2mks

A gland is a cell or group of cells that secretes a particular chemical substance for use in the body or for discharge into the surroundings.

Secretions are chemical substances, synthesized by glandular cells. They are released from the gland in response to an appropriate stimulus.

57.Write the role of Fibers of Zonula and ciliary muscles in eye? 3

Fibers of zonula held the lens in place. They change the shape of lens by exerting an outward tension on perimeter. Attached with fibers of zonula are the ciliary muscles

They contract to pull fibers of zonula, which flatten the lens. This focuses distant objects on retina. When they relax, lens becomes rounded and near object is focused.

58.Describe Hodgkin cycle in detail? 3

The number of Na+ channels open at any instant depends on Vm as well as on time i.e. the phase

of action potential. Thus changes in the conductance of sodium ions (gNa) occur as a function of Vm and time and reflect the behavior of thousands of Na+ channels, each one opening and closing during depolarization in accord with certain principles. During an AP, Na+ channels respond to an initial depolarization by opening, allowing Na+to enter the cell, which further depolarize the membrane. This depolarization causes more channels to open, allowing still more Na+to enter the cell and triggering an explosive, regenerative event. This relationship between membrane potential and sodium conductance is termed as the Hodgkin cycle and represents a type of positive-feedback system.

A stimulus causes local depolarization of the membrane. Na+ channels open in response to initial depolarization. This increases Na+ conductance that allows Na+ to enter the cell, which cause further depolarization of membrane and decrease in Vm. This relationship between Na+ conductance (gNa) and Vm represents a type of positive feedback system and is termed as the Hodgkin cycle.

It results in triggering of an action potential.

59.Which hormones are involved in the regulation of water and ions? 3

These hormones include:

- Antidiuretic hormone (ADH)
- Aldosterone
- Atrial natriuretic Hormone
- Calcitonin
- Parathormone

60.Messiner corpuscales and paccin corpuscales difference?3

Touch receptors in ridges of fingertips are Meissner's Corpuscles. They have encapsulated nerve endings. To receive pressure stimulus, Pacinian corpuscles are situated deep in skin, in the limbs they receive vibrations.

61.Basic components of homeostasis and example of homeostasis, Its mechanism (5max)

Examples of Homeostasis:

Homeostatic regulation of water • Homeostatic regulation of temperature • Regulation of pH • Regulation of glucose concentration • Regulation of osmotic pressure • Regulation of oxygen level • Regulation of ion conc.

Mechanism
Variables

Set point

Living control systems VS physical control systems

Three components: Receptors, Control centre, effectors

Example

• Temperature control system in air conditioners and water heating geysers VS Endothermic animals

62.Name glandular cells of thyriod and name their hormones too. (3mqx)

It contains five types of glandular cells that synthesize and secrete six hormones

1-Somatotropes produce growth hormone
(GH)

2-Corticotropes produce adrenocorticotropic hormone
(ACTH)

3-Thyrotropes
Produce thyroid-stimulating hormone (TSH)

4-Gonadotropes
Produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH)

5-Lactotropes
Produce prolactin (PRL)

63.Exocirne gland with 2 examples atleast (5max)

Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body. The fluid secretions may be proteins (enzymes) or mucous or both.

Examples:
Salivary glands produce saliva that is delivered to the oral cavity through submandibular and parotid ducts.

Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.

Lacrimal glands produce tears that is delivered through lacrimal duct on the surface of eye.

Mammary glands produce milk that is delivered through lactiferous ducts to the nipples

Androgens secreted by adrenal cortex 2 marks
Androgens secreted by adrenal cortex include:

1. Testosterone
2. Dihydrotestosterone
3. Androstenedione
4. Dehydroepiandrosterone (DHEA).

64.Effects of neuro-modulation 3 marks

- Development of complex behavioral patterns e.g. happiness, exploration, revenge, reward, greed.
- Processes of thinking, cognition, planning, learning and memory
- Behavioral problems: mood swings, sleep disturbances, feelings of stress, anxiety, anger and depression.

65. 2 examples of interoceptive receptors 3 marks

These include:
Receptors of vestibular system that monitor orientation of the body

Thermo-receptors that keep track of thermal state of the body

Chemo-receptors that keep track of chemical state of the body Proprio-ceptors that monitor position of the muscles and joints Nociceptors perceive the sensation of pain.
Pacinian corpuscles receive pressure stimulus.

66.Name and explain hair cell types 5 marks

Inner hair cells (IHC)

There are about 3,500 IHCs arranged in a straight line or wide U. They are true sensory cells which send impulses via auditory nerve.

Outer hair cells (OHC)

About 12,000-25,000 OHCs in mammals, they are arranged in three or four rows forming a characteristic W shape.

Outer hair cells have both sensory and motor elements. They contribute to hearing sensitivity and frequency selectivity

67.Hormones of ovary

LH stimulates production of estrogen and progesterone by ovary.

68.Explain photoreceptor

Photoreceptors can detect and generate response to a single photon of light. Photoreceptors possess light-sensitive carotenoid pigments retinal and 3dehydroretinal. Carotenoids are associated with opsin proteins to form rhodopsins. Rhodopsins absorb photons of light energy and produce a generator potential

69.Hair cilia types

Many cilia project from the apical end of each cell (reason of naming). • Cilia are of two types:

• Kinocilium • Stereocilia

A hair cell has a single kinocilium. • Kinocilium has a "9 + 2" arrangement of internal microtubules (similar to other motile cilia). • Each hair cell has 20-300nonmotile stereocilia. • Stereocilia are structurally and developmentally distinct from the kinocilium. • They are not formed of microtubules. • They are formed of actin filaments. • The stereocilia are arranged in order of increasing length from one side of the cell to the other.

70.Diseases by hypothyroidism

It is represented by two types of diseases: • Cretinism • Goiter

Cretinism results from iodine deficiency during early stages of development Inadequate production of thyroxine in adults leads to goiter.

71.active potential phases Stigma and ocelli 3

Eyespot or Stigma

Simplest photoreceptive structure. Found in some protozoa e.g. euglena. It is a bright red colored organelle and has carotenoid pigments. It gives a sense of light and dark. Helps in phototaxis.

Eyecups or Ocelli

Multi-cellular photoreceptive structure that consists of a cuplike depression containing photoreceptor cells; Found in cnidarians and flatworms e.g. Planaria.

Cannot form image, provides the animal a sense of direction only

72Functions of T3 and T4 and Hyperthyroidism 10. marks

These hormones have diverse effects on the physiology of virtually all tissues of the body. Their roles can be grouped into two categories:

1. Metabolic roles
2. Developmental roles

Metabolic Roles of T3 & T4

They stimulate cellular respiration, oxygen consumption and metabolic rate and, in turn, heat production. (important in thermoregulation).

They sensitize some tissues to epinephrine that helps to maintain normal blood pressure, heart rate and muscle tone.

They also promote normal motility of gastrointestinal tract.

They also regulate reproductive functions.

Developmental Roles of T3 & T4

They significantly affect the development and maturation of vertebrate animals.

They are involved in the normal functioning of bone-forming cells and the branching of nerve cells during embryonic development of the brain.

They control metamorphosis of a tadpole larva into adult frog.

Developmental effects of growth hormone occur only in the presence of thyroid hormones.

Hyperthyroidism

Excessive secretion of thyroid hormone is known as hyperthyroidism.

The most common form of hyperthyroidism is Graves' disease. It is an autoimmune disorder in

which antibodies that mimic TSH bind to the receptor for TSH and cause sustained thyroxine

production.

It leads to high body temperature, profuse sweating, weight loss, irritability, high blood pressure

and protruding eyes (exophthalmia).

Three use of crops 3 marks

In some animals, esophagus contains a sac-like expanded section, the crop, which is used to store food before digestion.
Crop is generally found in those animals that feed infrequently. It allows quantities of food to be stored for digestion at a later time Many birds e.g. chicken.
Leeches, which feed very infrequently, with weeks or months between feeding periods. In some animals crops are also used to ferment or digest foods for purposes other than normal digestive process. For example some parent birds prepare food in this way to be regurgitated for their nestlings e.g. pigeon's milk.

Role of nervous system in *regulating blood pressure 5 marks*

Role of Nervous System in Maintaining Blood Pressure
Nervous control serves to maintain arterial pressure by:
Adjusting resistance to blood flow in the peripheral circulation
controlling the number of capillaries open at any moment^
The nervous control of capillary flow ensures that if arterial pressure falls, blood flow to the gut, liver, and muscles is reduced to maintain flow to thaibraittand heart.
Nervous Innervations of Arterioles

Most arterioles are innervated by sympathetic nerves, which release norepinephrine at their endings.
Some arterioles are innervated by parasympathetic nerves, which release acetylcholine at their endings.
Sympathetic Nerve Stimulation
Stimulation of sympathetic nerves causes release of norepinephrine.
Binding of the norepinephrine to a-adrenoreceptors in the smooth muscle of arterioles causes vasoconstriction and therefore a decrease in diameter of the arterioles.
This decrease in diameter causes an increase in resistance to flow, thus reducing blood flow through that capillary bed.

34. Pose vertebrate have lungs

All vertebrate animals that live on land have lungs. When we breathe in, the muscle below the rib cage (called the diaphragm) is pulled down, and air gets sucked into the rib cage, filling the lungs. Blood cells circulating through tiny blood vessels near the lungs pick up oxygen and carry it around the body to the sites of respiration. Air is then forced out of the lungs as the diaphragm bows upwards

Baroreceptors 3 marks

Baroreceptors

The baroreceptor system is a simple and rapidly acting control mechanism that contributes to the regulation of arterial blood pressure

Baroreceptors in Vertebrates

Baroreceptors are widely distributed in the arterial system of vertebrates.
The central cardiovascular system of amphibia, reptiles and mammals has unmyelinated baroreceptors.
These unmyelinated baroreceptors only respond to pressures above normal, irritialrng reflexes that reduce arterial blood pressure and thus protect the animal from damaging increases in blood pressure.

Three ways of survival in cellular freezing. 3 marks

Such animals employ three methods to survive freezing environments:

- Ice-nucleating Agents (INAs)
- Cryoprotectants (antifreeze substances)
- Supercooling

38. What is core temperature

The core temperatures are maintained nearly constant between 37°C and 38°C in mammals and about 40°C in birds.

Panting in mammals 3 marks

Panting
Panting a method of cooling used by many mammals, birds and some reptiles involves use of respiratory system to loss heat.
During panting, mammals inhale through the nose and exhale through the mouth, exposing the tongue and other mouth structures to encourage further water evaporation and therefore heat loss.
It also involves increase in breathing rate and depth (hyperventilation).
Heat is carried away in exhalant air because the dimensions of the mouth are such that exhalant air retains the heat absorbed in the lungs.
In contrast, nasal passages and their vascularization are effective in retaining both water and body heat.

White *note on types of stomach vertebrates*

Types of Stomach in Vertebrates
Based on the number of chambers, stomachs are classified as:
Monogastric stomachs
Digastric stomachs

Monogastric Stomachs

A monogastric stomach consists of a single strong muscular tube or sac. Carnivorous and omnivorous vertebrates have a monogastric stomach. *J/*

Digastric Stomachs

Digastric stomachs are multichambered and are found in ruminant mammals of order Artiodactyla e.g. deer, giraffe, sheep, cattle, camel, llama etc.
These stomachs have four chambers^which are separated into two divisions.
The first division consists of fermentation chambers called rumen and reticulum
The second division comprises the omasum and abomasum (true stomach).

4 *steps of gas transport system 5 marks*

A gas-transfer system in animals takes in oxygen from the environment takes it to the issues through blood and brings CO2 from tissues, transporting it to the gas exchange surface and removes it by diffusion.

Such a system involves four basic steps:

- Breathing movements, which assure a continual supply of air or water to the respiratory surface (e.g., lungs or gills)
- Diffusion of 02 and C02 across the respiratory epithelium
- Bulk transport of gases by the blood
- Diffusion of 02 and C02 across capillary walls between blood and mitochondria in tissue cells

Apoenzyme and function

Apoenzymes- An enzyme that requires a cofactor but does not have one bound. An apoenzyme is an inactive enzyme^activajfion of the enzyme occurs upon binding of an organic or inorganic cofactor. The term holoen^yrneds bsed to refer to the apoenzyme bound to a cofactor.
Or Apoenzymes refers to inactive form of enzyme which activates upon the binding of a cofactor.

F unction;

It is responsible for reaction

Torpor with example 5 marks

Torpor

Small endotherms can not maintain their high rates of metabolic rates when they are not feeding during periods of inactivity.During those periods, some of these animals enter a state of torpor, in which body temperature and metabolic rate decrease.
The lower basal metabolism lowers the rate of conversion of energy stores into body heat.
Thus, it is generally advantageous to allow body temperature to decrease during periods of nonfeeding and inactivity.
As the animal comes out of torpor and becomes active, it undergoes a burst of metabolic activity, through shiverinjSpr oxidation of brown-fat and its body temperature rises.Daily torpor is practiced by many terrestrial birds. The hummingbird is a classic example, allowing its body temperature to fall from a daytime level of about 40°C to a nighttime level as low as 13°C.
Several species of small mammals also undergo torpor (e.g., shrews).

Write three type of blood vessels 3 marks

network of blood vessels

- arterioles,
- capillaries
- venules

and a network of lymph vessels, the largest of which is the central lacteal. Bloodvessels
Three types of blood vessels carry blood during circulation:
Arteries which carry blood away from the heart to organs and tissues.
Veins which return blood to the heart.
Capillaries which have thin walls formed of only a single layer of endothelial cells and permit exchange of materials between blood and tissues.